Training–Rollout Mismatch in Low-Precision LLM Reinforcement Learning: A Survey

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Abstract: Low-precision rollout can accelerate reinforcement learning for large language models, but the sampler generating responses and the learner computing updates may assign different probabilities to the same token history. Recent studies mitigate the resulting risks to estimation and optimization through quantization-aware training, importance weighting, and adaptive update control. We systematically organize these approaches in an intervention-based taxonomy with two complementary families. *Numerical design* aligns learner and sampler execution or preserves numerical fidelity and update visibility under quantization. *Learning under residual mismatch* corrects gradient estimation or controls the influence and acceptance of policy updates. The component-level view reveals how methods with different numerical targets and information requirements can complement one another. We analyze how discrepancies propagate through policy optimization and synthesize evidence on stability, policy quality, and total training cost. Smaller probability gaps can coexist with weaker learning, making sustained policy improvement central to evaluating mitigation. We identify future directions in long-term learning, efficient mitigation, method composition, and fair quality–cost comparison.

Keywords: quantized rollout; low-precision reinforcement learning; training–inference mismatch; importance sampling; quantization-aware training.

Resources: https://github.com/ZunhaiSu/awesome-low-precision-rl

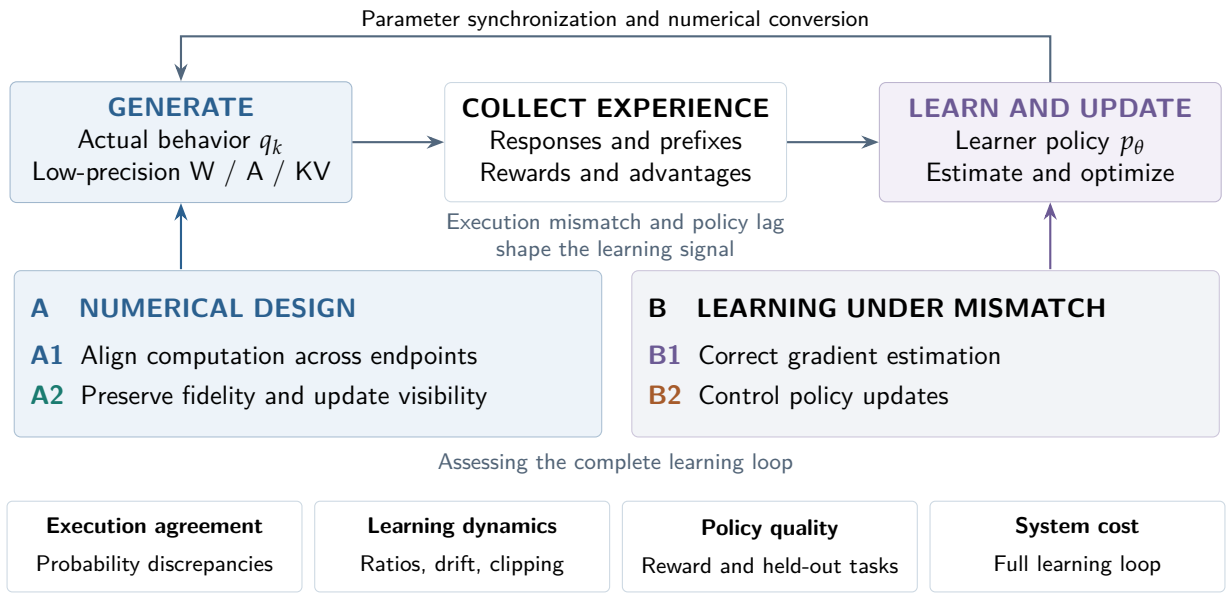


Figure 1: Survey at a glance. A changes numerical computation in the learner, sampler, or both; B corrects estimation or controls updates under residual mismatch. The two families can be combined. W, A, and KV denote weights, activations, and key–value cache.

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Reading pathways

Foundations	Methods	Evidence
Policy roles, execution mismatch, and its effects on learning.	Numerical interfaces, estimator correction, and update control.	Within-study findings, system costs, and open problems.
Sections 2–3	Sections 4–5	Sections 6–7

Appendix A: surveyed papers · Appendices B–C: sources and assumptions · Appendix D: terminology

Appendix E: mismatch measurements across core studies

1. Introduction and Scope

1.1. Why Low-Precision Rollouts Complicate Policy Optimization

Response generation can account for a substantial share of language-model RL training cost, particularly with long rollouts. Low precision can reduce arithmetic and memory costs [4, 12], but the training benefit depends on how the resulting experience supports policy improvement. Faster generation and more efficient learning are coupled objectives.

The sampler generates responses, while the learner recomputes token probabilities to form policy updates. Even with synchronized parameters, their kernels, numerical representations, routing decisions, or cached states can produce different probabilities for the same token history [1, 12]. Quantization can enlarge this training–inference mismatch (TIM), altering both the sampled experience and the weights or clipping decisions that determine its influence on learning. Updates then change the policy, its numerical behavior, and the histories sampled in later iterations. Numerical discrepancy therefore evolves with the learning loop.

Recent approaches act on different parts of this interface. Jet-RL coordinates precision flow across the two execution paths, FlashRL reweights behavior samples, and TRIAGE controls update contributions using mismatch direction [1, 3, 12]. Numerical agreement, stable optimization, and useful policy improvement are related but distinct outcomes. A correction can preserve learning under unequal policies, while early stability can conceal stagnation or later deterioration.

Understanding these approaches requires connecting each discrepancy to the mechanism used to address it and the evidence of learning benefit. Figure 2 organizes interventions into numerical design (A) and learning under residual mismatch (B). The classification applies to components, allowing their roles within combined systems to be understood separately. Appendix A maps the surveyed works to components and discussion locations.

1.2. Scope and Related Work

This survey focuses on numerical and execution mismatch in LLM reinforcement learning, especially quantized rollout and related low-precision training configurations. It covers a higher-precision learner paired with a lower-precision sampler, as well as low precision on both sides. Weight, activation, and key–value (KV) cache quantization affect different computational interfaces. Policy staleness enters where parameter changes interact with those interfaces.

The review covers nineteen core method and system studies, nine supporting formulation, theory, or general mismatch studies, and one exploration boundary case. Roles describe their use in the synthesis: MiniRL develops the formulation and also evaluates FP8 rollout [14]. Appendix B describes source selection and version handling.

Broader surveys cover low-precision training and reasoning-model RL [33, 35]. Generate–Filter–Control–Replay organizes rollout pipelines by their lifecycle functions [34]; the rollout-efficiency survey organizes techniques by mechanism and bottleneck, including composition and efficiency reporting [36]. Our focus is how numerical execution differences propagate into estimation, update decisions, and subsequent learning. Numerical alignment, correction, and cache design are examined through this shared interface [3, 6, 10, 12, 17].

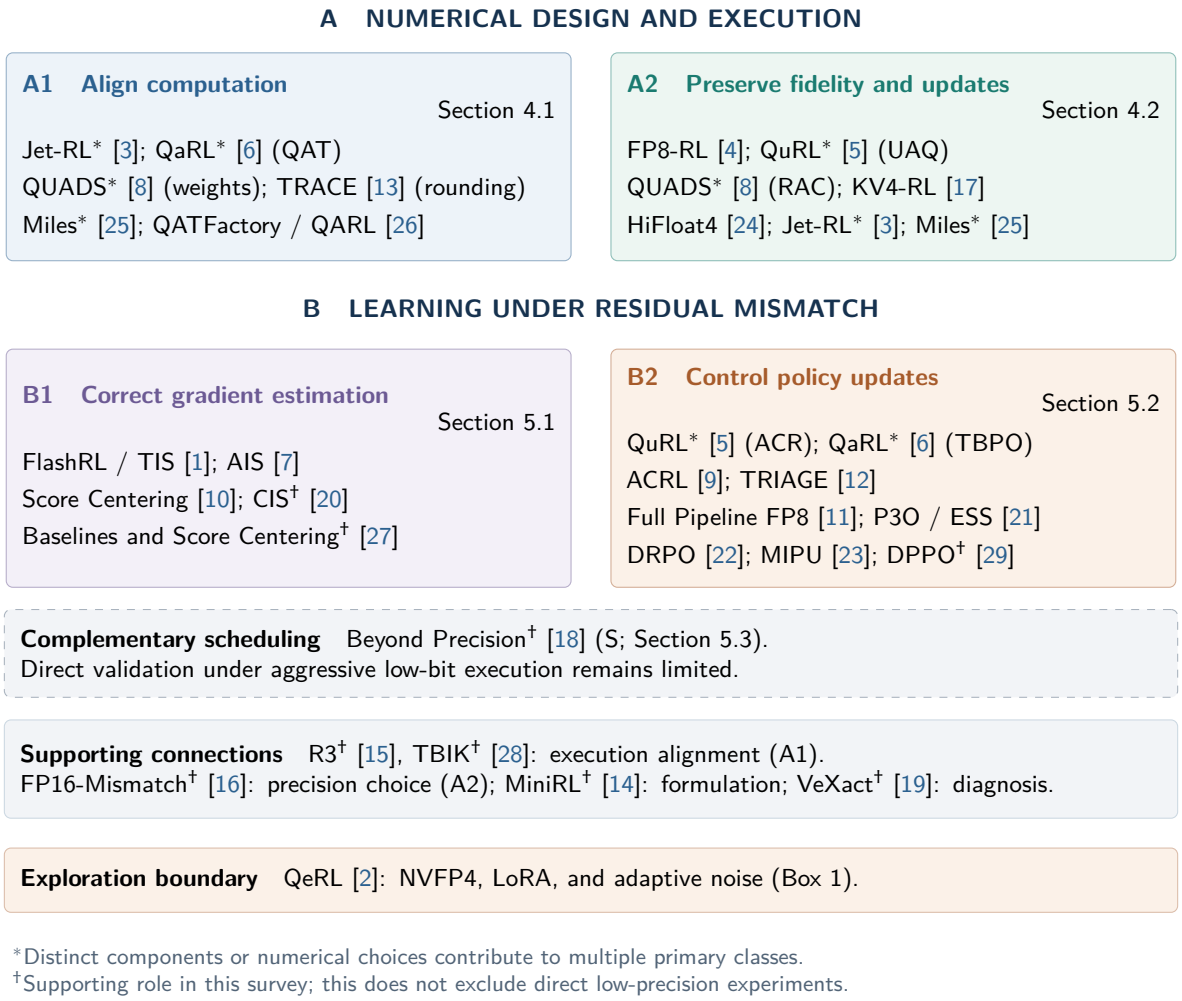


Figure 2: Component-level taxonomy of all twenty-nine focal studies. The four primary classes describe direct interventions. A work can occupy multiple classes through its components or numerical choices. Scheduling, supporting connections, and the QeRL exploration boundary are shown separately. Numerical regimes remain independent comparison axes. Appendix A gives the complete section mapping.

1.3. Perspective and Contributions

Distinguishing learner, behavior, proximal, and reference policies gives a common formulation of execution discrepancy and parameter lag. It clarifies the ratios and assumptions underlying estimator correction and update constraints.

The taxonomy classifies components by intervention target: computation alignment, fidelity and update visibility, estimator correction, and update control. Precision format, W/A/KV target, architecture, and collection protocol describe tested settings; granularity and information requirements distinguish components within a class.

The synthesis relates numerical agreement to estimation, updates, policy quality, and total cost. Diverging outcomes, residual errors, and the duration of supporting evidence guide future work on sustained learning, efficient mitigation, and quality–cost evaluation.

2. Background and Problem Formulation

2.1. Learner, Behavior, Proximal, and Reference Policies

Let x be a prompt, $y_{1:T}$ a response, and $h_t = (x, y_{<t})$ its prefix. A policy is specified by both a parameter state and the computation used to evaluate it. Write $p_\theta(a \mid h)$ for the current learner policy and $p_k(a \mid h)$ for the same learner-side path at sampling snapshot θ_k.

The behavior policy $q_k(a \mid h)$ is the distribution that actually sampled the token. It includes the rollout engine, numerical representation, and sampling transformation. Temperature and vocabulary truncation therefore belong to its definition: a raw-model log-probability returned by an engine need not equal the probability after decoding transformations.

The proximal policy p_{prox} defines the update ratio; the reference policy p_{ref} defines the regularization target. A frozen pre-update learner may supply the proximal policy, and an earlier checkpoint the KL reference. Neither necessarily generated the tokens. PPO and GRPO use these roles [30, 31]; mismatch-aware formulations explicitly distinguish them from behavior [1, 14].

A running GRPO update. Consider a prompt for which the sampler generates G responses. A task-dependent scorer assigns reward R_i to response i; in RL with verifiable rewards, a correctness checker can supply this score without a learned reward model. A common group-relative advantage is $A_i = (R_i - \bar{R})/(\sigma_R + \epsilon_A)$, where $\bar{R}$ and σ_R are the group mean and standard deviation, and $\epsilon_A > 0$ stabilizes the normalization. For binary rewards $(1, 1, 0, 0)$, the advantages are approximately $(1, 1, -1, -1)$. The advantage is held fixed during optimization and is typically shared by the valid tokens of the response. Some variants omit standard-deviation normalization [31].

At the learning stage, a representative clipped objective on this fixed group is

$$\begin{aligned} \widehat{J}(\theta) &= \frac{1}{G}\sum_{i=1}^{G}\frac{1}{T_i}\sum_{t=1}^{T_i}\left[\ell(r_{i,t}, A_i) - \beta \mathcal{K}_{i,t}(\theta)\right], \\ \ell(r, A) &= \min\{rA, \text{clip}(r, 1-\epsilon, 1+\epsilon)A\}. \end{aligned} \tag{1}$$

Here T_i counts valid response tokens, $r_{i,t} = p_\theta(y_{i,t} \mid h_{i,t})/p_{\text{prox}}(y_{i,t} \mid h_{i,t})$, ϵ sets the clipping interval, and β controls optional reference regularization. The conditional KL is $\mathcal{K}_{i,t} = D_{\text{KL}}(p_\theta(\cdot \mid h_{i,t}) \| p_{\text{ref}}(\cdot \mid h_{i,t}))$; implementations can use sampled estimators. Maximizing the surrogate encourages positive-advantage actions and discourages negative-advantage actions. Clipping limits incentives for large changes relative to the proximal policy; reference KL provides a separate anchor.

After the update, weights are synchronized and converted for the next rollout. With matched execution and decoding, the pre-update learner can also be the behavior policy: $q = p_{\text{prox}}$. Quantized rollout breaks this identification even before an update. At $p_\theta = p_{\text{prox}}$, the proximal ratio is one, but p_{prox}/q can already differ from one. Section 5.1 shows how an explicit behavior correction can be separated from the update ratio. Numerical design acts on these computations, estimator correction on the sampled experience, and update control on the learning step.

MIPU targets improvement in the inference policy obtained after conversion and synchronization, since it supplies the next experience [23].

Table 1: Policies and quantities used throughout the survey. Execution paths are part of policy identity.

Symbol	Meaning
p_θ	Current learner policy under its specified execution path.
p_k	Learner-side policy evaluated at sampling snapshot θ_k.
q_k	Actual behavior policy that sampled the token.
p_{prox}	Policy used to define the proximal update ratio.
p_{ref}	Policy used for KL regularization; not the behavior denominator.
Δ_k	Same-snapshot log-probability difference, $\log p_k - \log q_k$.
d_θ	Current learner–behavior gap, $\log p_\theta - \log q_k$.
A_t	Advantage or other policy-gradient learning signal.
$s_\theta(a,h)$	Score vector, $\nabla_\theta \log p_\theta(a \mid h)$.
$c_t, \bar{c}_t$	Behavior-to-proximal correction and its truncated or masked form.
$r_t(\theta)$	Proximal update ratio, $p_\theta / p_{\text{prox}}$.
$W_\theta(y \mid x)$	Complete-response likelihood ratio, P_θ / Q.
N, sg	Valid-token count and stop-gradient operator in Equation (10).

2.2. Execution Mismatch versus Policy Staleness

At a fixed prefix and token with positive relevant probabilities,

$$\frac{p_\theta(a \mid h)}{q_k(a \mid h)} = \underbrace{\left(\frac{p_\theta(a \mid h)}{p_k(a \mid h)}\right)}_{\text{parameter change / lag}} \times \underbrace{\left(\frac{p_k(a \mid h)}{q_k(a \mid h)}\right)}_{\substack{\text{same-snapshot} \\ \text{execution / decoding}}}. \tag{2}$$

The first factor captures parameter change since collection, including asynchronous version lag; the factorization requires no independence assumption. The second compares execution at the sampling snapshot. It isolates numerical differences when decoding is matched and also includes decoding differences when temperature or top-p filtering changes. The intermediate policy p_k is a conceptual reference and need not be retained by the implementation. In partial rollout, a response can span collection boundaries, so $k = k(t)$ may vary across tokens. If caches survive synchronization, retained entries encode earlier computations. Their provenance records the versions and numerical paths that produced them; a current checkpoint identifier alone does not specify this behavior state. The notation q_k abbreviates that stateful rollout path.

Figure 3 separates these snapshot comparisons from the two policy roles used by the loss. A proximal anchor and a KL reference can be chosen independently of the actual sampler.

At the sampling snapshot, the parameter-change factor is one, yet execution mismatch can remain. Conversely, matched execution and decoding make the second factor one without removing policy lag. Numerical alignment and staleness correction address different factors of the same ratio.

Importance ratios require the behavior policy to assign positive probability wherever the target does. Vocabulary truncation can violate this support condition even though every observed token has a defined denominator. At fixed h,

$$\mathbb{E}_{a \sim q_k(\cdot \mid h)}\left[\frac{p_\theta(a \mid h)}{q_k(a \mid h)} f(a,h)\right] = \mathbb{E}_{a \sim p_\theta(\cdot \mid h)}[f(a,h)] \tag{3}$$

holds under support and integrability assumptions. It corrects the action distribution at a fixed prefix while leaving the distribution over prefixes unchanged.

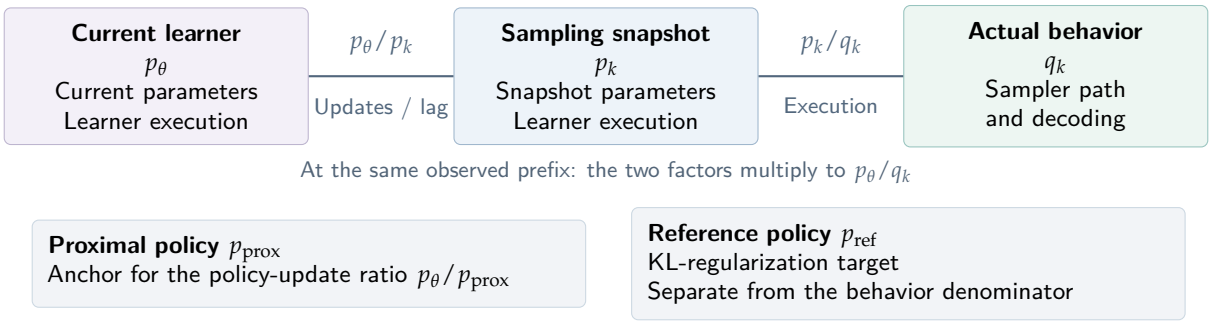


Figure 3: Policy identities and loss roles. Horizontal links denote the ratio comparisons in Equation (2), rather than data flow. Execution differences isolate numerical mismatch when decoding is matched. The lower cards identify separate roles in optimization.

The sampler can therefore visit histories with different frequencies from the target policy. Figure 13 illustrates this distinction, and Appendix C gives the full-response identities. In group-relative RL, changing the response distribution also changes the groups used to compute advantages [10].

2.3. A Component-Level Taxonomy of Interventions

We classify components by their *intervention target*: the computation or learning rule they change. The *source* of mismatch explains why probabilities differ; the *experimental setting* specifies precision, adaptation, and collection. Neither alone determines the method class.

A: numerical design intervenes in the computations that produce mismatch. A1 targets agreement across learner and sampler paths. A2 targets fidelity within a numerical path or the visibility of parameter changes after quantization. These components can act on the learner, the sampler, or both. **B: learning under residual mismatch** intervenes in how disagreement affects learning from collected samples. B1 corrects gradient estimation; B2 constrains, selects, or accepts policy updates. B can preserve useful learning without reducing the policy gap (Table 2).

Classification follows purpose. B1 derives correction from a specified expectation or gradient; B2 constrains, selects, or accepts updates. Both can weight a loss and affect estimation, while truncation can bias a B1 estimator. DPPO and DRPO change trust-region geometry (B2), while Score Centering corrects a conditional score expectation (B1) [10, 22, 29].

Several systems combine classes. QaRL pairs aligned QAT (A1) with TBPO (B2); QuRL pairs update-aware quantization (A2) with adaptive clipping (B2); QUADS pairs weight alignment (A1) with activation compensation (A2) [5, 6, 8]. Adopted corrections such as TIS are distinguished from each system's contribution. Optimization scheduling is a cross-family complement, denoted S and discussed separately in Section 5.3.

Table 2: Intervention targets and decision questions. A1 targets cross-path agreement; A2 preserves numerical information and update visibility. B1 targets an expectation; B2 targets an update decision. S is complementary scheduling. Figure 2 maps the studies.

Class	Decision question	What changes	Key dependency
A1	Can the two execution paths agree more closely?	Operators, representations, or realized numerical decisions.	Correspondence between computations.
A2	Is useful information preserved at low precision?	Scales, residuals, update resolution, or protected states.	Identification of sensitive information.
B1	How should behavior samples estimate a target gradient?	Importance weights or conditional score expectations.	Target distribution and available behavior information.
B2	Which updates should influence the next policy?	Constraints, contribution weights, or candidate acceptance.	Learning signal and control criterion.
S	When and how strongly should learning proceed?	Complementary optimization schedule.	A signal for adaptation over time.

2.4. Numerical Regimes and Contribution Roles

Quantization maps values to a discrete set using a specified format, scale, and rounding rule. Weight quantization changes parameters that can be shared within a sampling version. Activation quantization acts on values recomputed during each forward pass. KV quantization acts on attention state retained across token positions. These targets differ in when their errors arise and how long those errors remain in the computation.

Post-training quantization converts a learned model without adaptation through the target quantizer. Quantization-aware training (QAT) models quantization during learning. Fake quantization quantizes and dequantizes operands while matrix multiplications can remain at higher precision; a straight-through estimator supplies surrogate derivatives through rounding, as in QUADS [8]. Native kernels perform the specified low-precision arithmetic directly. QAT specifies adaptation; speed and backward, master-weight, and optimizer precision must be reported separately.

QuRL adapts to quantized rollout without an explicit learner-side QAT path: the sampler is repeatedly quantized while a higher-precision learner trains on its experience [5]. Low-rank or full-parameter updates, native or simulated arithmetic, architecture, and collection protocol specify the setting independently of the intervention target.

3. How Execution Mismatch Affects Learning

3.1. Sources and Propagation of Execution Discrepancy

Scales, rounding, accumulation, kernels, batch shape, and cached state can alter probabilities at the same nominal precision. FP16-based analysis examines precision selection [16]; R3 isolates routing differences that can occur without quantization [15].

Reduction order can change even when precision is fixed. TBIK coordinates reduction trees across tensor-parallel sizes, providing a matched-execution reference [28]. Under low precision, HiFloat4's ablations link degraded learning to outlier-driven underflow in rollout activations [24].

Shared-prefix probes compare conditional policies on fixed histories. Free-running comparisons also capture changed trajectories, where small errors can redirect generation.

Quantization can hide an update within one cell or amplify a small change across a boundary. QuRL seeks to make updates visible to the sampler; TRACE reduces inconsistent rounding [5, 13]. Both effects depend on the values' positions relative to quantization boundaries.

Backward arithmetic can change an update even when forward policies agree. Jet-RL specifies backward precision, and Miles offers a dequantized backward path to protect gradient fidelity [3, 25]. Interpreting low-precision failures therefore requires both policy-gap diagnostics and the learner's derivative computation.

Figure 4 shows how discrepancy changes sample influence, induces score drift, and alters update selection. An aggregate gap does not distinguish these mechanisms.

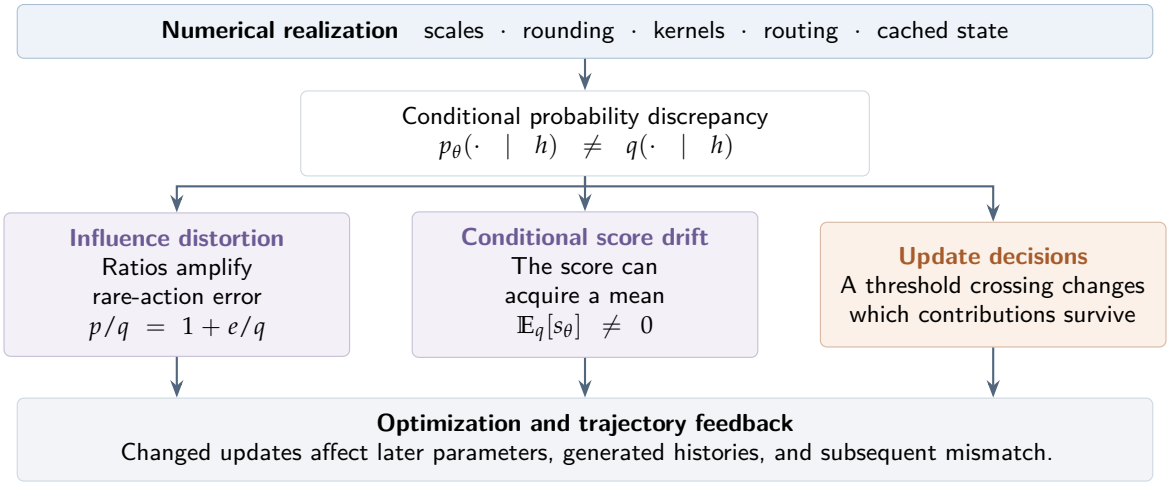


Figure 4: Optimization effects of execution discrepancy: relative influence [1], conditional score drift [10], and update decisions [11, 12]. The effects can interact through numerical and distributional feedback. Arrows indicate possible propagation under the stated policy and estimator assumptions.

3.2. Distorted Estimation and Update Decisions

Three mechanisms connect disagreement to optimization: relative sample influence, conditional score drift, and update decisions. Their effects depend on the policy, learning signal, and objective; absolute probability error alone does not determine the consequence.

Relative influence. Writing $p = q + e$ gives $p/q = 1 + e/q$: the same absolute perturbation has greater relative influence when q is small. Tail and confidence-conditioned statistics expose structure hidden by a mean absolute error; CIS uses confidence in its correction [20].

DPPO distinguishes relative ratios from moved probability mass [29]. A large ratio on a rare token can move little mass, while a moderate ratio on a common token can move much more. Ratio magnitude alone does not specify an appropriate update constraint.

Conditional score drift. Estimation can also acquire a systematic drift. For a normalized differentiable learner distribution, $\mathbb{E}_{a\sim p_\theta}[s_\theta(a,h)] = 0$. Under behavior sampling, the corresponding expectation need not vanish. At fixed h,

$$\mathbb{E}_q[As_\theta] = \mathrm{Cov}_q(A, s_\theta) + \mathbb{E}_q[A]\mathbb{E}_q[s_\theta]. \tag{4}$$

The expectations are conditional on the prefix and include the sampled action and any random continuation used to form A. Centering advantages across a prompt group need not center them at each prefix. Score Centering uses this distinction to motivate an additive correction [10]: it removes a first-moment drift term while leaving the behavior distribution itself unchanged.

Update decisions. Mismatch additionally affects decisions made by a surrogate objective. In a clipped policy-gradient objective, a small ratio perturbation can cross a branch boundary and remove a token's gradient contribution. The full-pipeline FP8 study isolates a failure involving excessive clipping of negative-advantage tokens [11]. This is a different intervention target from merely decreasing the variance of a sampled importance weight.

The sign dependence can be seen without assuming a particular quantizer. Let $\ell(r, A) = \min\{rA, \mathrm{clip}(r, 1-\epsilon, 1+\epsilon)A\}$, with $0 < \epsilon < 1$ and fixed A. Away from the two boundaries,

$$\frac{\partial \ell}{\partial r} = \begin{cases} A\,\mathbf{1}\{r < 1+\epsilon\}, & A > 0, \\ A\,\mathbf{1}\{r > 1-\epsilon\}, & A < 0, \\ 0, & A = 0. \end{cases} \tag{5}$$

Clipping depends on the advantage sign and the ratio being clipped. If the objective clips the combined learner–behavior ratio, $r = 0.7$ with $\epsilon = 0.2$ suppresses a negative-advantage token even at the sampling snapshot. This illustrates a different objective from the decoupled surrogate in Section 5.1.

Figure 5 illustrates the advantage-dependent clipping branches.

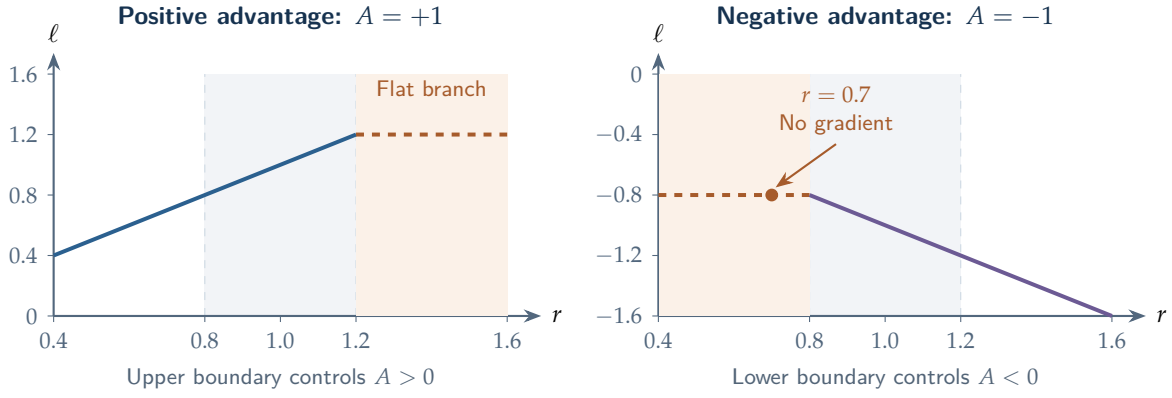


Figure 5: Signed clipping in Equation (5), with $\epsilon = 0.2$ and $A = \pm 1$. The gray interval is $[0.8, 1.2]$; amber segments are flat. At $r = 0.7$, a negative-advantage token contributes no gradient through this objective. Constructed example of the PPO surrogate with the specified ratio [30].

3.3. Feedback across Optimization Steps

The learning loop couples two forms of feedback. Numerical feedback arises when updates change values relative to quantization boundaries, scale statistics, and the sampler's ability to express parameter changes [4, 5]. Distributional feedback arises when the changed sampler visits different histories, producing different rewards, group advantages, and subsequent training data [10]. A fixed-batch discrepancy captures neither evolution by itself.

The direction of an update matters as well as the gap's magnitude. Locally, reducing a learner probability already below the behavior probability can enlarge the log gap. ACRL and TRIAGE use directional information to control updates [9, 12]. In a shared-parameter network, one token's gradient also changes other probabilities, so this local sign argument alone cannot guarantee stability.

Beyond Precision responds to this feedback with response-length-triggered learning-rate decay [18]. Its effectiveness under aggressive FP4 execution remains to be established.

3.4. Diagnosing and Measuring Mismatch

Studies measure computational disagreement, estimator reliability, update decisions, and learning outcomes (Table 3). Each answers a different question. Interpretation requires the compared policies, conditioning, and aggregation, recorded for all nineteen core studies in Appendix E.

Policy comparison and conditioning. A *controlled numerical probe* fixes parameters and prefixes to compare execution paths. An *online diagnostic* compares the current learner with recorded behavior and may include policy lag (Equation (2)). Figure 6 contrasts the two. Raw, decoded, and recomputed probabilities remain distinct; VeXact supplies a matched-execution reference [19].

KV4-RL computes full-vocabulary $D_{\mathrm{KL}}(p_{\mathrm{BF16}} \| q_{\mathrm{quant}})$ at identical BF16-generated prefixes from the same checkpoint, before temperature or truncation [17]. The probe isolates conditional KV distortion and excludes trajectory selection. Conditioning also differs within TRIAGE: its 30B characterization replays sampled MoE routes, whereas its 4B run measures the ordinary learner–sampler gap. Their raw magnitudes are therefore not directly comparable [12].

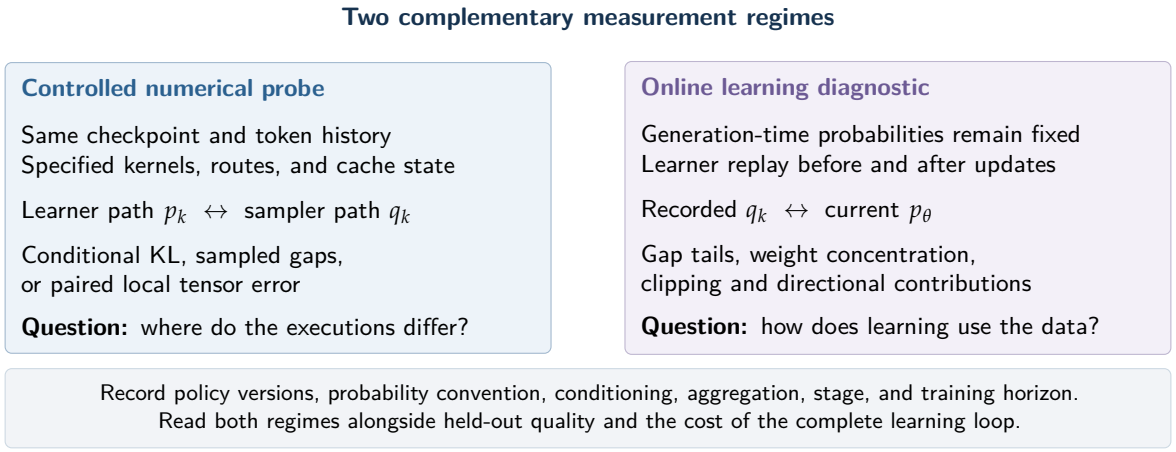


Figure 6: Controlled probes and online diagnostics target different comparisons [12, 17, 20]. Execution attribution depends on the factors held fixed; online gaps can include policy lag.

Probability gaps and KL answer different questions. For a specified pair of normalized conditional policies p, q, let $d_t = \log p(a_t \mid h_t) - \log q(a_t \mid h_t)$ on sampled actions $a_t \sim q(\cdot \mid h_t)$. Common summaries include

$$G_{\text{abs}} = \frac{1}{N}\sum_t |d_t|, \qquad \widehat{K}_1 = -\frac{1}{N}\sum_t d_t, \qquad \widehat{K}_3 = \frac{1}{N}\sum_t (e^{d_t} - d_t - 1). \tag{6}$$

Here N counts valid measured tokens, and G_{abs} removes the gap's sign. With common support and finite expectations, both KL estimators target conditional $D_{\text{KL}}(q\|p)$ averaged over the observed prefixes. In finite samples, $\widehat{K}_1$ can be negative, while every $\widehat{K}_3$ summand is nonnegative. These sampled estimates differ from reverse-direction $D_{\text{KL}}(p\|q)$, full-vocabulary evaluation, and KL to a regularization reference.

TRIAGE and AIS report mean absolute log gaps; TRACE reports maxima and minima; QUADS tracks a maximum gap; QuRL reports behavior-to-proximal KL and a maximum ratio; MIPU uses sampled-token K3 [5, 7, 8, 12, 13, 23]. Extremes can expose errors hidden by a mean, but also depend on population size and composition. Their magnitudes are comparable only under matched definitions, populations, and reductions.

Table 3: Complementary levels of mismatch diagnosis. Each level answers a different question; outcomes require evidence connecting them to the preceding mechanisms.

Question	Representative measurements	Interpretation boundary
Where do computations disagree?	Shared-prefix KL; sampled-action gaps and tails; paired activation error [8, 12, 13, 17].	Specify checkpoint, prefix, routing, probability convention, and tensor location.
How reliable is the estimator?	Raw or capped weight concentration; ESS; conditional score mean; variance amplification [7, 10, 21].	Concentration and drift are different effects. A theoretical estimand is not automatically a logged statistic.
Which updates change?	Sign-separated clipping; directional mass and segment localization; candidate acceptance [11, 12, 23].	Requires the objective, fixed advantages, and the relevant update or synchronization stage.
Does learning remain useful?	Reward, held-out quality, entropy, failure time, and total cost.	Training outcomes assess consequences; they do not uniquely identify numerical mismatch.

Estimator and update diagnostics. Normalized empirical effective sample size (ESS) summarizes weight concentration:

$$\text{ESS}_{\text{norm}}(w) = \frac{(\sum_t w_t)^2}{N\sum_t w_t^2}. \tag{7}$$

P3O computes ESS from raw current-to-behavior ratios; AIS uses truncated weights [7, 21]. The values can therefore differ on identical data. Uniform nonzero weights give normalized ESS one regardless of their common value, so high ESS establishes neither calibration nor gradient fidelity. Score Centering addresses the conditional score mean instead (Section 5.1) [10].

The full-pipeline FP8 study uses BF16 shadow passes to identify tokens whose active clipping branch changes under FP8, separating the two advantage signs [11]. This counterfactual *over-clipping* is not an ordinary clipping fraction. TRIAGE measures directional mass and local concentration from signed gaps, then tests the indicator against realized updates [12]. These diagnostics assess how discrepancy changes an update; legitimate learning can still increase disagreement.

Measurement population and aggregation. Population and intervention stage condition every diagnostic. Token and response averages weight the same data differently; masking can lower a gap by selection, and capping can raise ESS by reducing its tail. Appendix E records the reductions, retained populations, and probability conventions needed to interpret these effects.

Training duration and delayed instability. Longer optimization and longer generated trajectories test different properties. To assess persistence across updates, report optimizer steps, rollout and learner tokens, and elapsed training cost, and retain late-stage measurements alongside the best checkpoint. Periodic paired probes separate execution error from parameter lag; online diagnostics and held-out quality track its learning consequences.

Time to failure requires a declared event and persistence window, such as a sustained quality loss assessed alongside estimator or update diagnostics. Independent training runs expose variation in its onset. A run that ends without the event bounds the observation window; it does not establish an unlimited failure time. Longer runs should also track retained learning contributions, since a controller can suppress instability by progressively suppressing learning.

A useful diagnostic connects a specified discrepancy to an estimator or update effect and tests its consequences for learning quality over a stated horizon and cost. Minimizing every online gap would also penalize legitimate policy movement.

Box 1. Quantization and exploration

QeRL combines low-precision execution, low-rank adaptation, and adaptive quantization noise, making exploration an explicit part of the training design [2]. It occupies a boundary of this survey: the numerical perturbation changes the search process as well as agreement between computations. The reported benefit concerns structured quantization noise under low-rank adaptation and depends on the induced sampling distribution and the learner's adaptation capacity.

The following sections examine remedies in the computations that produce disagreement (Section 4) and in the learning rules that use the resulting experience (Section 5).

4. Numerical Design and Execution Alignment

Numerical design can act on the learner, the sampler, or the interface between them. Alignment brings their computations closer at a shared interface; fidelity control preserves information that a low-precision representation would otherwise lose. The distinction matters because two approximate paths can agree without either becoming more accurate relative to a higher-precision model.

4.1. Aligning Learner and Rollout Computation

Alignment couples operator precision, quantized representations, or realized rounding outcomes. The coupled object determines the required coordination and residual discrepancy.

At the operator level, Jet-RL coordinates precision across the training and rollout computation graphs [3]. Corresponding forward operators share precision and quantization granularity. Linear-layer forward and backward GEMMs use FP8 inputs, while gradients transported between operators remain BF16. This precision flow extends beyond learner-side fake quantization.

QaRL combines rollout-aligned learner QAT (A1) with TBPO (B2) [6]. QAT matches the quantized representation; TBPO controls sequence-level updates on problematic low-bit responses.

QUADS treats weights and activations differently [8]. Synchronized weights can share a quantize–dequantize path, whereas activations are recomputed from engine-dependent inputs. The same coarse quantizer can then send nearby values to different codewords. QUADS fake-quantizes learner weights, retains BF16 learner activations, and compensates selected rollout activation errors. Its asymmetric-versus-symmetric ablation shows that quantizing the same operands on both sides need not improve agreement. Section 4.2 examines the activation compensation.

Representation alignment still leaves rounding outcomes dependent on the inputs to each path. TRACE therefore couples realized rounding decisions [13]. A shared FP4 format and scale can still send nearby learner and rollout activations to different codewords. TRACE records rollout-side quantization information and uses it to guide the learner's corresponding quantization site. With exact guidance and fixed rollout scales, it selects from the learner value's two neighboring codewords the one nearest the rollout codeword. Figure 7 illustrates why this can reduce cross-path discrepancy even when it slightly increases the learner's own quantization error.

The implementation approximates this reference using compact rollout records. By default, TRACE caches one low code bit and the scales from the latter half of the layers, pairing records by trajectory, token position, and quantization site. The reconstruction uses the current learner activation as side information: after normalization by the rollout scale, it selects the nearest E2M1 magnitude whose retained code bits match the record, taking the sign from the learner. The adjacent learner candidates are then compared with this reference. The record narrows a local choice; it cannot independently recover arbitrary rollout values. Guidance can be omitted in earlier layers; reconstruction depends on paired activations remaining locally adjacent.

TRACE's main rollout quantizes expert W/A and softmax-layer KV to NVFP4, retaining BF16 elsewhere. The learner uses guided fake quantization at matching sites, with BF16 matrix multiplication, attention, and backward computation. Routing replay and token-level correction address residual differences; unpaired sites and policy lag remain sources of error [13].

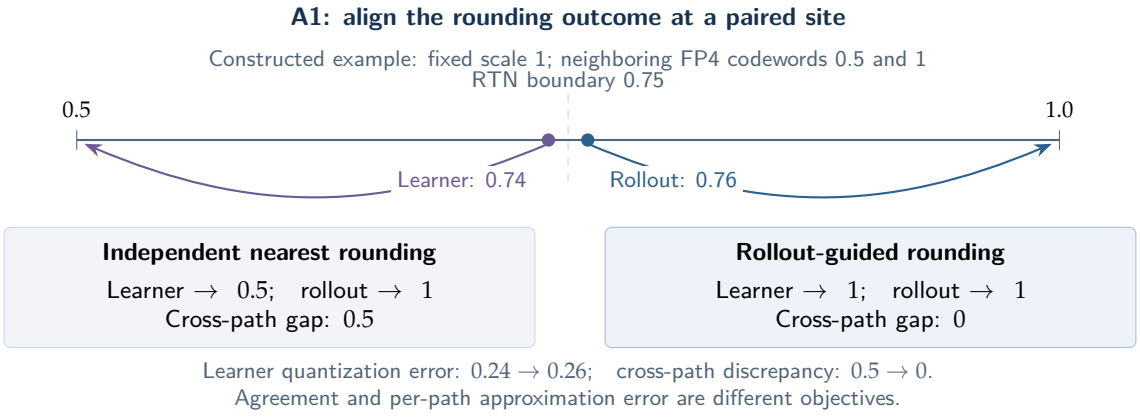


Figure 7: Rollout-guided rounding in a constructed fixed-scale E2M1 example. Agreement improves while learner approximation error increases slightly. Exact guidance applies at the paired site; TRACE's compact implementation approximates the reference [13].

Alignment must survive conversion and synchronization. Miles coordinates checkpoint conversion, learner forward, rollout kernels, and live export using shared quantization rules; its MXFP8 and

NVFP4 recipes include precision exceptions and are marked Beta [25]. QATFactory's QARL trains with BF16 quantization simulation, uses native NVFP4 rollout, and exports low-bit weights [26].

The shared numerical object determines the correspondence required between tokens, modules, and parameter versions. Routing replay aligns MoE choices [15]; TBIK aligns reduction order [28]. These interfaces leave other operations to be aligned or corrected separately.

4.2. Controlling Quantization Error and Preserving Parameter Updates

A2 preserves information within the numerical path through two mechanisms: retaining activation or state fidelity, and making parameter changes visible after quantization.

Preserving update visibility. Alignment alone does not ensure that the sampler can express the changes made by training. For an illustrative deterministic quantizer Q, an update can satisfy

$$Q(w + \delta w) = Q(w), \tag{8}$$

even when $\delta w \neq 0$. Visibility depends on update magnitude, distance to a quantization boundary, and scale behavior. QuRL's Update-Aware Quantization applies a one-time invariant rescaling before RL [5]. The identity $WX = (W/s)(sX)$ preserves the initial real-valued layer output while changing the magnitudes seen by quantization and optimization. Rescaling is chosen to improve update size relative to quantization noise. Reparameterization changes gradients and can alter later optimization. UAQ targets sampler update visibility; ACR controls the learner's clipping region (Section 5.2).

Figure 8 shows a smaller update crossing a boundary while a larger one stays within its cell. Changing the scale moves these boundaries.

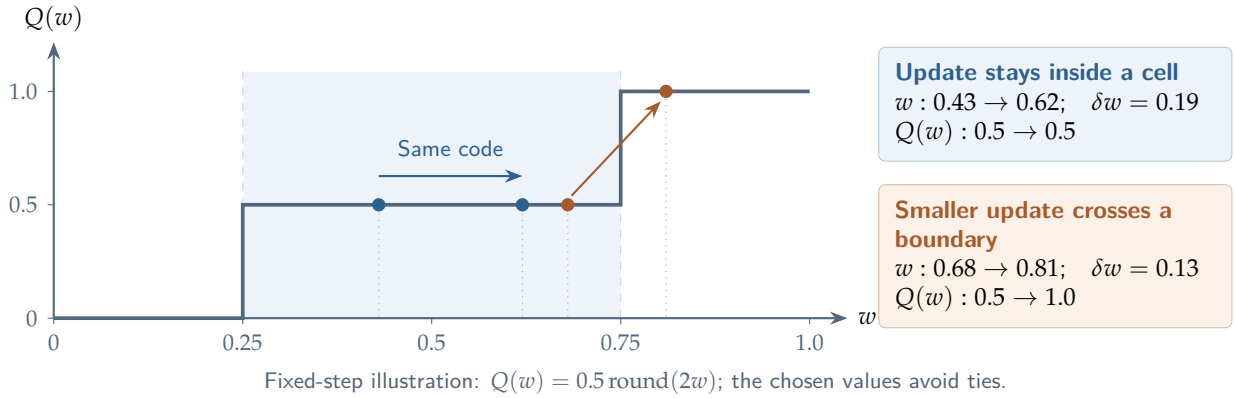


Figure 8: Constructed illustration of update visibility (Equation (8)). The blue update stays inside a quantization cell while the smaller amber update crosses a boundary. Changing scales moves the boundaries.

Preserving activation and state fidelity. QUADS complements learner-side weight alignment with Residual Activation Compensation, restoring selected high-error channels around native W4A4 rollout computation [8]. Compensation changes the sampler's activation approximation.

HiFloat4 / Rollout-ResQ corrects rollout activation underflow with sparse residuals [24]. Its learner–sampler precision comparisons isolate an error that raising learner precision alone does not resolve. These FP4 experiments simulate the arithmetic and do not measure native-FP4 acceleration.

FP8-RL maintains numerical fidelity as the policy changes through blockwise W8A8 rollout, scale recalibration for FP8 attention and KV execution, and dynamic weight synchronization [4]. It integrates these components with adopted TIS/MIS correction. Its v2 experiments also explore end-to-end FP8, so learner precision varies across the paper's experimental settings.

Miles retains selected tensors in BF16 and offers an optional dequantized backward path that trades training throughput for gradient stability without changing the quantized forward values [25]. These choices specify the fidelity of its forward and derivative computations separately.

KV4-RL uses INT4 KV with bounded INT8 protection for attention-sink and recent positions [17]. TIS remains fixed across cache-precision comparisons, isolating the benefit of changing the sampler computation. In colocated, quota-based partial collection, generation and optimization alternate; responses can span collection boundaries.

Update visibility preserves the effect of learning; activation compensation, recalibration, and cache protection preserve the information used to generate experience [4, 5, 17, 24]. Both must track a changing policy. Their runtime and coordination requirements differ from the learner-side quantizer correspondence required by QAT (Table 4).

4.3. Residual Mismatch and Limits of Alignment

Alignment compares learner and sampler computation at a specified interface, snapshot, and history. Fidelity measures approximation to a higher-precision reference. Shared quantizers can still disagree because their inputs or surrounding kernels differ.

Numerical alignment alone does not guarantee reliable estimation or policy improvement. A faithful sampler can still lag behind the learner, while agreement at observed prefixes leaves the history distribution changed by earlier sampling decisions. Learning-side correction and control address these residual effects on estimation and updates.

Maintaining a shared numerical interface requires information as well as arithmetic. Operator choices require coordination, QAT depends on matching quantizer semantics, and rounding guidance requires storage and correspondence with learner states [3, 6, 8, 13]. Fidelity control instead uses calibration, compensation, or selective execution to preserve sensitive information [4, 5, 17]. Table 4 compares these dependencies and the discrepancies each leaves unresolved.

Table 4: Numerical strategies: shared information, execution dependencies, and residual differences.

Strategy	Shared or protected object	Information dependency	Residual discrepancy
Precision flow	Corresponding operator precision and granularity	Coordinated numerical configuration	Engine and surrounding-kernel differences.
Aligned QAT	Learner representation corresponding to rollout	Matching quantizer semantics	Native versus simulated arithmetic; unmatched activations.
Guided rounding	Realized rollout quantization decisions	Stored and matched rollout guidance	Unshared modules and downstream propagation.
Fidelity control	Scales, update resolution, residuals, or protected states	Calibration, compensation, or selective execution	Other execution paths and policy staleness.

5. Learning under Residual Mismatch

Learning methods address the experience already collected. Estimator correction changes how behavior samples estimate a target gradient; update control restricts their influence on the next policy. Scheduling adjusts the scale of successive updates. These choices affect estimation bias and variance as well as learning progress and stability.

5.1. Correcting Gradient Estimation

Multiplicative correction and truncation. Equation (3) gives the conditional change of measure. A decoupled surrogate separates behavior correction from the proximal update ratio. For a frozen proximal policy p_{prox} and recorded behavior q, define

$$c_t = \frac{p_{\text{prox}}(a_t \mid h_t)}{q(a_t \mid h_t)}, \qquad r_t(\theta) = \frac{p_\theta(a_t \mid h_t)}{p_{\text{prox}}(a_t \mid h_t)}. \tag{9}$$

A representative decoupled surrogate is

$$\widehat{L}(\theta) = \frac{1}{N} \sum_t \text{sg}(\bar{c}_t) \min[r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t], \tag{10}$$

where $\bar{c}_t$ is a possibly truncated correction, advantages are held fixed, N counts valid sampled tokens, and sg prevents differentiation through the correction weight. Before truncation and clipping, $c_t r_t = p_\theta / q$ compares the current learner with the actual sampler. Bounding c_t limits collection-related correction; clipping r_t limits the incentive for policy change during the update. Upper truncation uses $\bar{c}_t = \min(c_t, C)$, whereas masking discards selected ratios. Response-level averaging and length normalization assign different relative weights, making the reduction rule part of the estimator.

FlashRL uses actual rollout probabilities for truncated importance sampling (TIS) [1]. MiniRL distinguishes behavior and proximal policies [14]. FP8-RL adopts token-level TIS and masked importance sampling (MIS), while KV4-RL retains TIS alongside cache protection [4, 17]. Truncation bounds influence but introduces bias.

Adapting the amount and location of correction. Adaptive importance sampling (AIS) makes correction strength depend on both necessity and reliability [7]. Its effective multiplier has the form $1 + \alpha(\bar{w} - 1)$: $\alpha = 0$ retains the uncorrected estimator, while $\alpha = 1$ applies the truncated importance weight $\bar{w}$. A mean absolute log gap indicates whether correction is needed; capped-weight ESS and advantage-weight variance amplification indicate whether it is reliable. AIS can therefore reduce correction strength under large mismatch when the weighted estimator becomes too concentrated or noisy. The adapted quantity is the estimator mixture.

Calibrated importance sampling (CIS) adjusts the truncation cap according to confidence. It represents cross-engine perturbation as a log-odds displacement, which can produce very different ratios for low- and high-probability actions [20]. Thus, AIS adapts the mixture strength, whereas CIS adapts the cap. CIS evaluates BF16 cross-engine mismatch; transfer to FP4 remains untested.

Additive score correction. Score Centering addresses a different object from the sampled-token importance weight [10]. Here the *score* is the parameter-gradient vector $s_\theta(a, h) = \nabla_\theta \log p_\theta(a \mid h)$,

not the reward assigned to an answer. Its mean is zero under the learner distribution, but can be nonzero under the actual behavior distribution. In Equation (4), that conditional mean produces an additional term proportional to the conditional mean advantage. Group-centered rewards need not remove it because prompt-level centering does not center the signal at every generated prefix.

Score Centering replaces the score at a prefix with

$$\widetilde{s}_\theta(a,h) = s_\theta(a,h) - \mathbb{E}_{b\sim q(\cdot|h)}[s_\theta(b,h)]. \tag{11}$$

With the exact expectation, its mean under q vanishes and the drift term in Equation (4) is removed [10]. The remaining covariance is still measured under q, so this is not full recovery of an on-policy gradient. The reported implementation uses sampler top-k log-probabilities and models the omitted tail using the learner distribution; its default is $k = 128$ [10]. The omitted mass therefore enters through a modeled tail rather than being discarded. The coefficients used to form the correction are detached, and its fidelity depends on that behavior reconstruction.

Figure 9 gives a one-parameter example. Changing the gradient's conditional mean does not require changing the sampler or forcing the learner probabilities to equal it. This separates a correction to the estimator from direct numerical alignment. When combined with importance weighting, the centered object must be the corresponding weighted score; an arbitrary weighted loss with the original unweighted center need not have the intended cancellation [10].

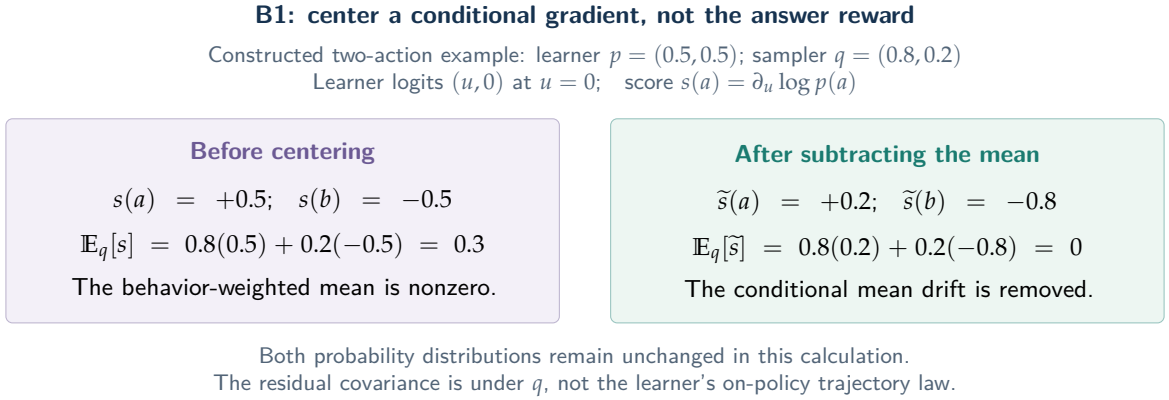


Figure 9: Constructed scalar-logit example of score centering [10]. Sampler probabilities weight the derivatives in Equation (11). Centering changes the conditional gradient mean while leaving both distributions fixed; practical top-k reconstruction adds an approximation.

Baselines and score centering interact differently at token and sequence level [27]. For the sequence-level regularized importance estimators studied there, the group-mean baseline already supplies the centering benefit. At token level, that baseline generally leaves bias under regularized ratios. Centering must match the estimand, weights, and reduction.

These estimators trade fidelity against variance and information cost. Truncation and masking control tails but bias estimates. TIS, AIS, and CIS use sampled-action probabilities, while Score Centering uses a reconstructed behavior law.

5.2. Controlling the Influence of Policy Updates

Update control has three descriptors: the decision rule, its control signal, and its application unit (Table 5). The rule can reshape constraints, select contributions, or accept a candidate policy. Signal

and application scales can differ: P3O uses batch statistics, and ACRL uses response statistics, to guide token contributions [9, 21].

Table 5: Update-control mechanisms within B2, with their decision signals and aggregation units.

Decision rule	Signal or constraint	Application unit	Components
Boundary placement	Mismatch-aware, trust-band, or reference-calibrated clipping.	Token or response.	QuRL ACR; QaRL TBPO; Calibrated Clipping.
Batch adaptation	ESS determines cap and regularization.	Batch statistics, applied through the objective.	P3O.
Constraint geometry	Divergence-based boundary or smooth regularizer.	Token-level probability shifts.	DPPO; DRPO.
Directional influence	Advantage and discrepancy determine retained or corrective contributions.	Tokens and segments within responses.	ACRL; TRIAGE.
Candidate acceptance	Assessment after inference-side synchronization.	Whole candidate policy.	MIPU.

Restoring or reshaping update constraints. QuRL's Adaptive Clipping Range (ACR) adjusts the permitted update region using proximal and behavior probabilities [5]. It addresses clipping distorted by mismatch already present at collection. UAQ (Section 4.2) addresses a different failure: the quantized sampler may not express the resulting weight change. QuRL therefore acts on both learning an update and transmitting it through quantization.

QaRL's TBPO constrains whole responses through negative-sample treatment and a sequence-level trust band [6]. Full-pipeline FP8 instead calibrates clipping against BF16: it matches the lower quantile on negative-advantage tokens, then adjusts the upper boundary to rebalance signed contributions [11]. Periodic reference passes amortize calibration of the update boundary.

Clipping statistics require the advantage sign, reduction, and comparison policy to be specified. An aggregate fraction cannot reveal which useful contributions a boundary removes.

Adapting control to batch reliability. P3O uses normalized effective sample size (ESS) of the batch's policy ratios to adapt a score-weight cap and off-policy regularization [21]. Its FP8-rollout experiment studies numerical mismatch directly, while broader comparisons show dependence on the model and collection setting. AIS adapts an estimator mixture; P3O adjusts update constraints.

Choosing the constraint geometry. DPPO introduces a divergence-based trust region [29]. DRPO retains Binary-TV geometry but replaces hard masking with a smooth regularizer that supplies corrective gradients outside the boundary [22]. DRPO evaluates FP8-rollout-only and end-to-end FP8 MoE settings, extending the available precision evidence for this update geometry.

Controlling disagreement through update direction. ACRL steers disagreement through advantage-guided token reweighting controlled by response-level discrepancy from a reference [9].

TRIAGE connects the advantage sign to the learner–sampler log gap [12]. Using $d_\theta = \log p_\theta - \log q_k$ in this survey's notation, the product $A_i d_{\theta,i,t}$ identifies the sign of a simplified token's own mismatch-energy contribution. Cross-token coupling and optimizer history can change the realized motion, so this is a local directional indicator. Both $A_i > 0, d_\theta > 0$ and $A_i < 0, d_\theta < 0$ are locally amplifying. TRIAGE intervenes selectively on the latter: increasing the probability of a reward-positive response can be legitimate learning even if its gap grows.

The gate first diagnoses 64-position segments of negative-advantage responses from their mean gap and deep negative tail. Within an eligible risky segment, only negative-gap tokens receive an attenuated raw weight $b_{i,t}$. The weighted response loss is normalized by the sum of its weights. Relative to ordinary response averaging, its effective multiplier is therefore $b_{i,t}/\bar{b}_i$, not simply $b_{i,t}$. Normalization reallocates contributions: for illustrative raw weights $(0.3, 1, 0.3, 1)$, the effective multipliers are approximately $(0.46, 1.54, 0.46, 1.54)$. If every weight is equal, the normalization cancels it. This controls relative influence, not the total gradient norm.

Repair targets positive-advantage responses with sufficiently negative mean gaps in 128-position segments. A one-sided pseudo-Huber penalty raises learner log probabilities. Its derivative bounds the additive coefficient, not the full gradient or every token's gap. All valid tokens in an active segment share that coefficient, including positive-gap tokens. Gating and repair thus address different signed contributions [12]. The tested recipe uses native W4A4 forward on both endpoints and BF16 KV. Applying it to KV4 remains an untested transfer.

Accepting whole updates in the inference policy. MIPU constructs candidates relative to the sampler, then assesses whether a synchronized candidate should become the next inference policy [23]. This separates candidate construction from acceptance after numerical conversion. The practical inference-gap proxy supports acceptance or rollback in the evaluated FP8 runs; a general guarantee of monotonic improvement would require stronger conditions.

Granularity governs localization and statistical reliability. Token or segment rules can target concentrated effects; batch aggregation can improve signal reliability; whole-policy acceptance assesses the synchronized result. These describe different decision scales, while clipping and regularization specify how a decision is enforced.

5.3. Adapting the Optimization Schedule

Beyond Precision adapts the learning-rate schedule to the evolution of optimization and mismatch [18]. Scheduling changes the scale of successive parameter updates, complementing numerical alignment and sample-level control. Direct evidence under aggressive quantization remains limited. Its value depends on preserving quality within the same budget, since smaller steps can delay progress as well as divergence.

5.4. Compatibility, Bias, and Failure Modes

Complementary methods address different residual errors. Weight alignment can leave activation error, while a more faithful sampler can still generate off-policy data. QaRL, QuRL, and QUADS test combinations addressing such residual effects [5, 6, 8].

Table 6: Studied component combinations, their intervention targets, and the residual effects addressed.

Study	Combined components	Mechanistic connection
QuRL [5]	Update-aware quantization (A2) and adaptive clipping (B2).	Make parameter changes visible to the sampler and reshape the learner's update constraint.
QaRL [6]	Aligned QAT (A1) and TBPO (B2).	Reduce representational disagreement, then constrain problematic response-level updates.
QUADS [8]	Weight QAT (A1) and residual activation compensation (A2).	Weight alignment leaves activation error for a separate numerical component.
KV4-RL [17]	Selective cache protection (A2) with retained TIS.	Improve the data-generating path while holding a bounded conditional correction in the recipe.
TRACE [13]	Guided QAT (A1), routing replay, and token-ratio correction.	Couple rounding outcomes while separately addressing route choices and residual off-policy effects.
Score Centering [10]	Centering composed with TIS or MIS (B1).	Center the corresponding weighted score; reconstruction and regularization determine the remaining bias.

Estimator compatibility depends on weighting, centering, masking, and reduction. Token- and sequence-level regularized estimators can require different centers [27]. Figure 10 distinguishes this estimation target from the contributions selected by update control.

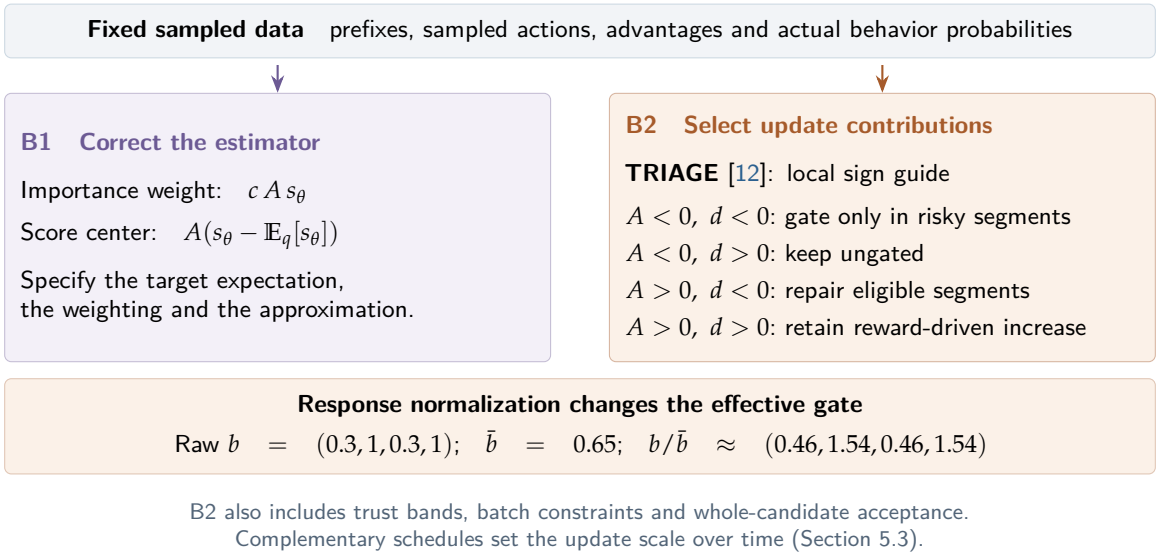


Figure 10: Estimator correction and update control. TRIAGE [12] uses $d = d_\theta$ with segment-level conditions. Repair acts on positive-advantage, negative-mean segments and includes all their valid tokens. Raw weights illustrate normalization, rather than a diagnosed segment; shared parameters limit token-wise predictions.

Score Centering removes a conditional score mean; TRIAGE selects contributions using advantage and log-gap signs. Their joint effect remains untested [10, 12].

Composition must preserve policy identities and estimator semantics. Corrections use the policy that generated each token, even when later numerical changes alter the sampler. An importance-weighted score needs its corresponding weighted center [10]; successive masks can suppress the same contributions. Guidance and reference passes also compete for memory, computation, and bandwidth. The combinations in Table 6 support their tested implementations; prospective combinations require compatible information, weighting, and reduction rules.

6. Evidence Synthesis and System Trade-offs

Published evidence connects numerical agreement, policy improvement, and total training cost under specific models, execution paths, evaluation protocols, and budgets.

6.1. Experimental Regimes

A higher-precision learner paired with a quantized sampler exposes differences between representations. Quantizing both sides can still leave different kernels, activations, and updates. KV quantization changes retained attention state; weight and activation quantization affect other computations. Low-rank and full-parameter adaptation have different capacities to accommodate these errors.

AIS evaluates diffusion-based LLaDA alongside autoregressive Qwen models [7]. The Qwen experiments follow the token-level formulation used here; the denoising results require likelihoods defined for their generation process.

Evaluation precision determines which policy is assessed. KV4-RL uses a common BF16 evaluator to compare learned parameters under a fixed execution path [17]. TRACE evaluates its FP4 policy, including the low-precision realization of those parameters [13]. These protocols can yield different quality assessments of the same checkpoint.

Architecture and collection protocol further condition conclusions. MoE execution introduces discrete route choices; cache-heavy long responses expose different runtime bottlenecks from shorter generations; partial collection permits a response to contain tokens sampled under different parameter versions. These differences affect both the mismatch mechanism and the cost of intervention.

Miles adds production-scale agentic system evidence, whereas QATFactory isolates a smaller NVFP4 QARL workload [25, 26]. Native low-precision arithmetic, simulated quantization, and BF16 execution on quantized operands are distinct regimes. In particular, HiFloat4 reports simulated FP4 RL rather than measured native-FP4 training [24].

Table 7 summarizes four evaluated learner and rollout configurations. Appendix B maps coverage across numerical, architectural, and collection regimes (Table 12).

6.2. Findings on Stability and Learning Quality

Component comparisons can test whether correcting a measured error improves learning. Figure 11 shows the baselines in TRACE and TRIAGE; Table 9 compares their findings with Score Centering and KV4-RL. Appendix E records the diagnostic practices of all nineteen core studies.

Table 7: Representative learner, rollout, and collection configurations. QAT specifies adaptation; forward, backward, and optimizer precision are separate attributes.

Study	Learner path	Rollout and collection setting
TRACE [13]	Guided fake-quantized forward; BF16 backward.	NVFP4 W4A4 on routed experts and KV4 in softmax attention; other modules BF16. Asynchronous collection.
TRIAGE [12]	Native NVFP4 W4A4 forward.	Native W4A4 forward; BF16 KV in the documented characterization. Synchronous math RL; no reported KV4 experiment.
Score Centering [10]	BF16 learner; distributional reconstruction supplies the correction.	Quantized-sampler stress tests include FP4 KV and INT8 sampler with INT4 KV. Policy-lag tests form a separate setting.
KV4-RL [17]	BF16 learner.	INT4 KV with bounded INT8 protection. Quota-based partial collection; generation and learning alternate. This is a cache-focused experiment.

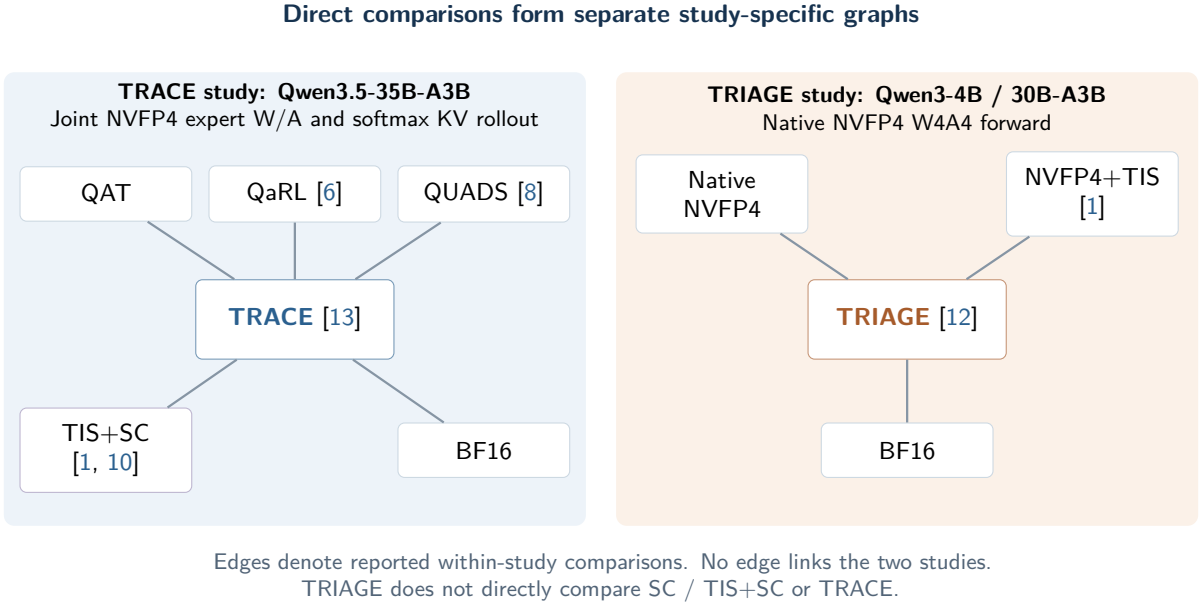


Figure 11: Study-specific comparison graphs for TRACE and TRIAGE [12, 13]. Each BF16 reference belongs to its study's execution setting. Table 9 gives outcomes and checkpoint conditions.

Numerical agreement and useful learning can diverge. Entropy, reward, and probability gaps describe different aspects of a training trajectory (Table 8). Section 3.4 defines mismatch measurements; their joint evolution reveals whether closer agreement accompanies useful learning.

TIS prevents aggregate-gap divergence in TRIAGE's 30B experiment but trails the full method in late reward and downstream scores [12]. Useful reward-driven updates can also increase signed gaps. Retained contributions and policy quality are therefore necessary to interpret gap control.

Controller signals require evaluation against independent learning outcomes. AIS uses a gap to adapt correction, KV4-RL measures cache distortion with conditional KL, and TRIAGE uses signed, localized gaps to design its controls [7, 12, 17]. Learning outcomes establish whether the intervention improves policy quality beyond its effect on the controlled metric.

Table 8: Reading learning trajectories. Definitions illustrate common conventions; the source policy and reduction determine the logged quantity.

Quantity	What it measures	What it does not establish
Policy entropy H	$-\sum_a p(a)\log p(a)$ for a specified next-token policy, subsequently aggregated over prefixes.	A lower value need not indicate better learning. A label such as *entropy loss* does not establish that entropy regularization is active.
Mean reward $\bar{R}$	$B^{-1}\sum_i R_i$ over B sampled responses. With strictly binary correctness reward, it is the sampled correctness rate.	It is not the group-normalized advantage, a held-out benchmark score, or an invariant scale across reward functions.
Mean absolute gap G	Magnitude of learner–behavior disagreement on valid sampled tokens; Equation (6).	It removes sign and local concentration; it is neither KL divergence nor proof of estimator fidelity or policy quality.

Durability depends on retained learning. Durability concerns whether an intervention remains effective as the policy and numerical state evolve. QuRL's disagreement after initial stability motivates joint attention to clipping and update visibility [5]. Adaptive correction, directional control, and score centering act on different parts of this feedback [7, 9, 10]. P3O's Qwen3-8B FP8 experiment links batch-adaptive control to preserved later performance over the evaluated horizon [21].

TRACE reports increasing discrepancy and later quality loss for conventional FP4 QAT in its joint W/A/KV setting [13]. TRIAGE explicitly leaves substantially longer training horizons unevaluated [12]. A demonstrated late failure and an unevaluated continuation support different conclusions. Stability claims are bounded by the observed horizon. Retained contributions and policy quality help distinguish sustained learning from stability through suppression.

Alignment depends on the shared numerical interface. QUADS' asymmetric weight QAT recovers much of the lost performance, and rollout activation compensation adds further gains; its symmetric-QAT comparison shows that quantizing more learner operands can worsen agreement [8]. HiFloat4's simulated ablations also locate residual error in rollout activations [24]. These results support matching the relevant numerical interface, which differs across shared precision flow, representations, and rounding decisions [3, 6, 13]. FP8-RL and QATFactory further show why calibration, conversion, and the evaluated representation belong to that interface [4, 26].

Learning effects depend on update decisions. The full-pipeline FP8 study links excessive lower-side clipping of negative-advantage tokens to pathological training dynamics; TRIAGE selects interventions using local, directional information [11, 12]. DRPO preserves corrective gradients outside its trust region; MIPU assesses a whole candidate after conversion to the inference policy [22, 23]. These controls act at different locations and stages, which aggregate clipping fractions or mean gaps cannot distinguish. TRIAGE's routing-conditioned MoE analysis also finds that a population-level directional trend need not predict every token's motion. Checkpoint selection further affects the comparison: the full-pipeline FP8 study chooses its main GRPO endpoints by best average performance over 500 steps [11], whereas TRIAGE uses the final scheduled step for stable arms. Similar endpoint scores can summarize different learning trajectories.

Table 9: Within-study evidence with mechanistic interpretations. The controlled setting, checkpoint rule, and evaluation protocol bound each conclusion.

Study	Controlled comparison	Reported result	Mechanistic reading and scope
TRACE [13] 35B MoE	Joint NVFP4 W/A/KV rollout; TIS+SC uses cap 2 and top-128 reconstruction.	Four-task mean: TRACE 75.3, TIS+SC 74.1, BF16 74.9.	Supports the guided-alignment recipe in this joint regime. SC uses its best mean checkpoint through 400 steps; task-wise gains are mixed and aggregate uncertainty is unreported.
TRIAGE [12] 30B MoE	BF16, native NVFP4, TIS, and full gating/repair recipe. Stable arms use step 1700.	Five-task mean: TRIAGE 70.96, TIS 63.18, BF16 72.41.	Supports direction-aware stabilization beyond TIS in native W4A4. SC/TIS+SC, TRACE, and KV4 are outside this comparison.
Score Centering [10] 30B MoE	Correction rules share the objective; quantization and lag are tested separately.	With FP4 KV, SC and TIS remain stable; stronger INT8/INT4 stress separates them further.	Supports estimator correction under unequal policies. Curves report training accuracy; shared-objective controls do not reproduce every baseline recipe.
KV4-RL [17] 8B dense	Cache precision varies with TIS retained; all learned policies are evaluated in BF16.	At 400 updates, six-task mean: 52.8 versus BF16's 52.4; cumulative full-iteration speedup 1.27×.	Supports selective cache protection with TIS retained. Common BF16 evaluation assesses learned parameters; evaluation-seed variation differs from training-run variation.

TRACE tests guided alignment against QAT and TIS+SC in joint low-precision execution; TRIAGE tests recovery beyond TIS in native W4A4. These different baseline sets and precision regimes preclude a direct TRACE–TRIAGE ranking [12, 13].

Residual errors motivate component combinations. Ablations identify which residual error makes a second component useful. QUADS separates weight alignment from activation compensation, QuRL combines update visibility with clipping control, and KV4-RL changes cache fidelity while retaining TIS [5, 8, 17]. These results support the tested combinations; extending them requires identifying the residual error in the new setting.

6.3. Efficiency Gains and Implementation Costs

Computational benefit depends on the complete learning loop. Under serialized execution, iteration time is

$$T_{\text{iter}} = T_{\text{rollout}} + T_{\text{learner}} + T_{\text{sync}} + T_{\text{quant/calib}} + T_{\text{other}}, \tag{12}$$

with synchronization and quantization/calibration accounted for separately. Accelerating one phase increases the relative cost of the others. Overlapped execution instead depends on the critical path, for which the additive model in Equation (12) does not apply.

For fixed workload and remaining phases, rollout fraction f, rollout speedup s, and added work hT_0 give $T/T_0 = (1-f) + f/s + h$ (Figure 12). Overlap, response-length changes, or altered concurrency require a different model.

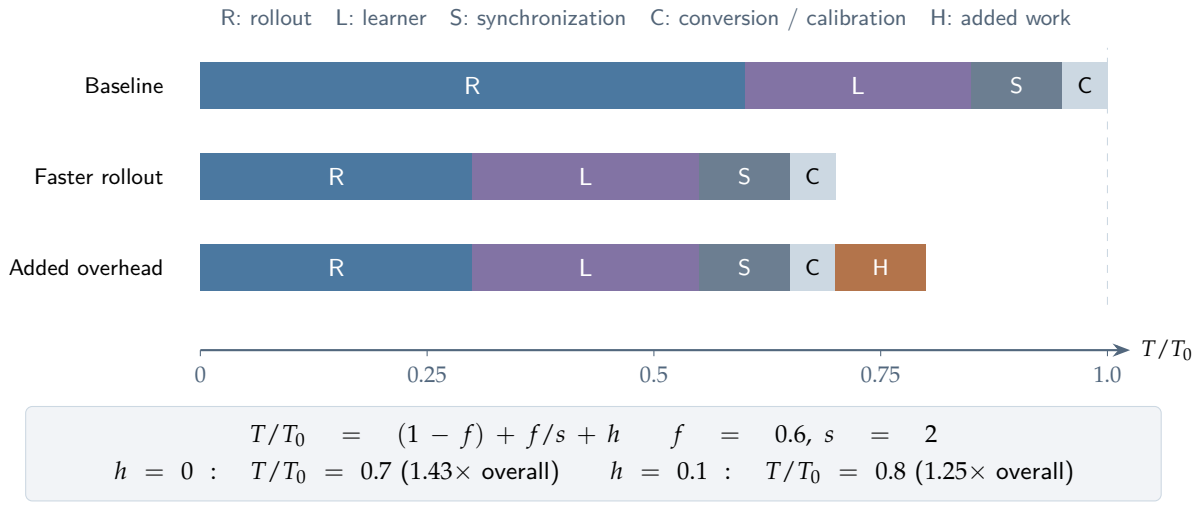


Figure 12: Serialized timing with a faster rollout phase and added overhead. Doubling rollout speed reduces iteration time from T_0 to $0.7T_0$ when rollout accounts for 60% of the baseline. Added work of $0.1T_0$ reduces the overall speedup from 1.43× to 1.25×. Constructed illustration under serialized execution.

Mitigation redistributes this cost across the loop. Scale estimation and conversion add work in FP8-RL [4]; reference calibration adds learner passes; guided rounding requires storage, transport, and state matching; selective precision increases work in protected regions. Table 10 connects these costs to each method's information requirements.

Table 10: Information and execution dependencies that determine implementation cost.

Approach	Information or execution dependency	Consequence for implementation
TIS / AIS / CIS	Behavior probabilities and learner/proximal probabilities	Stored probabilities must include the sampling transformations.
Score Centering	A behavior-weighted score expectation, exact or approximated	Exact evaluation requires distributional information; reconstruction introduces approximation.
Aligned QAT / shared precision	Matching quantization and conversion rules across endpoints	Precision exceptions, kernels, and weight export form part of the configuration.
Guided rounding	Realized rollout-side quantization information	Guidance requires storage, transport, and matching to learner states.
Calibrated Clipping [11]	Higher-precision reference ratios at calibration points	Reference passes add training computation, amortized across calibration intervals.
Batch-adaptive control [21]	Policy-ratio statistics aggregated over the batch	Reduction rules and the sampled batch determine the reliability estimate.
Inference-policy acceptance [23]	Assessment of the synchronized candidate	Candidate evaluation and possible rollback add work beyond learner updates.
Selective KV precision	Low-bit cache execution with protected positions	Protected positions add work whose share depends on the context workload.

Practical scope depends jointly on the numerical setting, available information, and the cost of acquiring it. P3O requires batch-level ratio statistics, while MIPU's acceptance step adds inference-side assessment and possible rollback [21, 23]. Miles and QATFactory emphasize conversion and inference-ready weight transfer as part of the execution contract [25, 26]. TBIK's coordinated reductions can carry a kernel-performance cost [28]. Rollout-ResQ's sparse correction is designed for efficient hardware execution, but its simulated experiments do not measure a native FP4 speedup [24].

The useful quantization target depends on the workload as well as the format. QUADS targets native W4A4 expert computation, whereas KV4-RL studies how sustained concurrency and long contexts increase the value of reducing KV processing [8, 17]. Their settings motivate different numerical targets; a bit-width comparison alone does not identify the dominant cost.

Throughput depends on response length and concurrency. Shorter responses can reduce generation time without improving per-token execution; longer responses can increase work despite faster kernels. Memory savings can raise concurrency and change the kernel workload. PagedAttention illustrates the effects of cache management and scheduling on serving throughput [32]; RL efficiency additionally depends on the resulting learning trajectory.

KV4-RL reports 1.80× cumulative rollout speedup and 1.27× cumulative full-iteration speedup over BF16 rollout across 400 Qwen3-8B-Base iterations [17]. The difference reflects the cost of the remaining phases. Its common-BF16 evaluation separately compares learned-policy quality, with both results tied to the evaluated collection workload.

Comparable quality–cost evidence reports three views [17, 36]. *Quality at a fixed budget* compares policies at matched total resources under a specified evaluator. *Cost to a target quality* includes failure to reach the declared target within budget. *Extended-budget trajectories* assess persistence of benefit. Accounting includes extra learner passes, response-length changes, and required updates. Overlapped execution also requires measuring guidance traffic on the critical path.

6.4. Evidence Limits and Coverage Gaps

Strong evidence isolates a mechanism within its execution setting. Mathematical reasoning is common; interactive tasks and hardware or kernel transfer have fewer controlled comparisons. Native arithmetic, quantization simulation, and BF16 execution on quantized values remain distinct.

Learning comparisons need a training horizon, budget, checkpoint rule, and uncertainty unit alongside the metric definition in Section 3.4. Best-checkpoint scores can hide later deterioration. Repeated decoding measures evaluation variation; independent RL runs expose training variation. Smoothing alone quantifies neither.

Separate W4A4 and KV4 studies do not establish their joint behavior; asynchronous demonstrations do not isolate every precision–lag interaction. Scheduling needs direct low-bit validation, while diffusion needs generation-specific likelihoods before the autoregressive analysis can transfer.

7. Open Problems

The central challenge is to preserve policy improvement as numerical execution, sampled states, and training budgets evolve. The reviewed evidence points to five connected directions.

7.1. Predicting Harmful Mismatch and Sustaining Learning

Predictive diagnostics should identify lost learning before held-out quality deteriorates. Ratio tails, conditional score drift, clipping errors, and segment concentration offer mechanism-specific signals. Their value depends on predicting subsequent learning loss beyond ordinary policy movement or changes in the selected data. Paired probes and fixed-batch audits isolate execution and update effects; continued training tests whether those effects predict the timing and severity of deterioration.

Durability should be assessed as the policy, activation statistics, and sampled states evolve. Extended runs can distinguish sustained improvement, postponed deterioration, and stagnation under increasingly restrictive control. Adaptive recalibration, guidance allocation, and update control should be judged by the learning progress they preserve and their adaptation cost. Successful mitigation can support continued learning while residual probability gaps remain.

ACRL identifies a specific durability question: its reference discrepancy is fixed at initialization, and substantially longer training may require a moving baseline [9]. Score Centering notes that its severe-mismatch, short-sequence experiments proxy longer training under milder mismatch [10]. Establishing when such stress tests predict extended-run behavior would make them more informative without treating them as substitutes for that evidence.

7.2. Composing Numerical and Learning Interventions

Numerical and learning interventions should address complementary residual errors. Weight alignment can leave activation error; correction can alter the need for restrictive control. Score Centering leaves a reward–score covariance under behavior sampling, motivating importance-sampling composition under severe staleness [10]. Beyond the studied combinations (Table 6), guided rounding with centering and cache protection with directional control remain open. Joint design could allocate precision, guidance, and correction under one budget. Ablations should separate error removal, retained contributions, and added cost from retuning.

7.3. Reducing Information and Control Costs

TRACE records low code bits and scales for selected layers [13]. Further compression should preserve the information guiding useful updates. Adaptive layer, token, or phase selection could be assessed under a budget covering acquisition, storage, transfer, reconstruction, and synchronization.

TRIAGE combines segment tests, token gates, normalization, and repair. Its proposed adaptation to mismatch magnitude, locality, and persistence [12] motivates simpler controllers. Evaluation should relate retained learning to overhead and tuning effort at matched budgets.

QaRL proposes token-level approximations to its sequence objective; MIPU identifies cheaper candidate assessment as an open choice [6, 23].

7.4. Modeling Stateful and Asynchronous Behavior

TRIAGE leaves asynchronous validation open; KV4-RL identifies disaggregated systems, other attention architectures, and multi-turn workloads as extensions [12, 17]. These settings introduce new discrepancy and state dependencies.

Responses can cross synchronization boundaries, and retained KV entries encode earlier computations. The modeling question is which token, version, and cache records suffice for correction and controlled comparison (Equation (2)), without retaining the full computation history.

Early perturbations change later prefixes beyond conditional token correction. Precision allocation should therefore follow learning benefit across layers, positions, and training phases. Multi-turn workloads extend this dependence to tool observations and state transitions.

7.5. Generalization and Quality–Cost Evaluation

Transfer depends on preserving each method's information requirements when execution or collection changes. AIS also leaves exploration under quantization unresolved [7]; ablations should distinguish exploration, smoothing, and update weighting. Component comparisons should share execution and the base objective; system comparisons should preserve essential design under matched resources. The quality–cost views (Section 6) should test transfer across architectures, collection protocols, and budgets, with comparison conditions explicit.

8. Conclusion

Low-precision RL couples numerical execution with estimation, updates, and future experience. Execution agreement, optimization stability, and policy improvement are distinct outcomes. Sample probabilities, conditional score expectations, and update decisions explain why an average policy gap alone cannot establish learning benefit.

Interventions should be assessed by the residual errors they address, their required information, and their application scale. These connections explain complementarity and its cost. Predictive diagnostics, efficient mitigation, and comparable evidence across training budgets support the practical goal: sustained policy quality at lower total training cost.

A. Surveyed Papers and Section Mapping

Table 11 maps the twenty-nine focal studies to regimes, components, and discussion locations. Years mark first public appearance; bibliography entries may cite later publications. Parentheses give additional discussion locations. Appendix B defines contribution roles.

Table 11: Surveyed papers: numerical regimes, method components, and section mapping. Conference and preprint versions count as one contribution.

Work / first year	Numerical regime	Component(s) and contribution	Discussion
Core method and system studies (19)			
FlashRL / TIS [1] 2025	Higher-precision learner; quantized rollout	**B1**: Recorded behavior probabilities and truncated correction.	**5.1** (1.1; 2.1)
Jet-RL [3] 2026	Unified FP8 learner and rollout paths	**A1; A2**: Unified FP8 precision flow.	**4.1** (6.2)
FP8-RL [4] 2026	W8A8 rollout; FP8 attention / KV; FP8 learner explored in v2	**A2; system**: W8A8 rollout, FP8 attention/KV, and stack integration.	**4.2** (5.1; 6.2; 6.3)
QuRL [5] 2026	INT8 / FP8 rollout	**B2; A2**: Adaptive clipping and update-aware quantization.	**5.2** (3.1; 4.2; 5.4; 6.2)
QaRL [6] 2026	Quantized MoE rollout; aligned learner QAT	**A1; B2**: Rollout-aligned QAT and trust-band updates.	**4.1** (5.2; 5.4; 6.2; 7.3)
P3O / ESS [21] 2026	BF16 learner / FP8 rollout; dense reasoning models	**B2**: Batch-adaptive ESS cap and regularization.	**5.2** (6.2; 6.3)
AIS [7] 2026	BF16 learner / FP8 rollout; autoregressive and diffusion models	**B1**: Adaptive importance-weighted estimator mixture.	**5.1** (6.1; 6.2; 7.5)
DRPO [22] 2026	BF16, FP8 rollout-only, and FP8 end-to-end MoE settings	**B2**: Smooth corrective divergence regularization.	**5.2** (6.2)
MIPU [23] 2026	BF16 learner / FP8 rollout; Qwen3-1.7B and 4B	**B2; formulation**: Inference-policy candidate construction and acceptance.	**5.2** (2.1; 6.2; 6.3; 7.3)
QUADS [8] 2026	NVFP4 MoE rollout; learner weight QAT	**A1; A2**: NVFP4 MoE weight alignment and activation compensation.	**4.1** (4.2; 5.4; 6.2)
ACRL [9] 2026	FP8 rollout; discrepancy-dependent updates	**B2**: Discrepancy-dependent update control.	**5.2** (3.3; 6.2; 7.1)
HiFloat4 / Rollout-ResQ [24] 2026	Simulated end-to-end HiF4 / MXFP4; rollout residual correction	**A2**: Sparse residual correction for FP4 rollout activations.	**4.2** (3.1; 6.2; 6.3)
Miles v0.1 [25] 2026	FP8 / MXFP8 / NVFP4 RL recipes; precision exceptions	**A1; A2; system**: Shared quantization, synchronization, and precision contracts.	**4.1** (4.2; 6.1; 6.3)
Score Centering [10] 2026	BF16 learner; FP4/INT4 KV stress and separate lag tests	**B1**: Additive behavior-weighted score correction.	**5.1** (2.2; 3.2; 5.4; 6.2; 7.1; 7.2)
Full Pipeline FP8 RL [11] 2026	FP8 learner and rollout; calibration against BF16	**B2; system**: FP8 learner/rollout and calibrated clipping.	**5.2** (3.2; 6.2; 6.3)

Continued on next page

Table 11 continued: Surveyed papers

Work / first year	Numerical regime	Component(s) and contribution	Discussion
QATFactory / QARL [26] 2026	Native NVFP4 rollout; simulated-quantization BF16 training	**A1; system**: Deployment-aligned QARL and inference-ready weight synchronization.	**4.1** (6.2; 6.3)
TRIAGE [12] 2026	Native W4A4 forward; documented BF16 KV; synchronous RL	**B2**: Direction-aware control for native NVFP4.	**5.2** (3.3; 6.2; 7.3; 7.4)
TRACE [13] 2026	NVFP4 expert W4A4 + softmax KV4; guided QAT, BF16 backward	**A1**: Rollout-guided FP4 rounding for MoE.	**4.1** (3.1; 6.2; 7.3)
KV4-RL [17] 2026	BF16 learner; INT4 KV with bounded INT8 protection	**A2; system**: INT4 KV with selective INT8 protection.	**4.2** (5.1; 6.2; 6.3; 7.4)
Supporting formulation, theory, and general TIM studies (9)			
R3 [15] 2025	MoE routing; general execution mismatch	**Support; A1**: MoE routing alignment.	**3.1** (4.1)
FP16-Mismatch [16] 2025	BF16 / FP16 numerical comparison	**Support; A2**: BF16/FP16 precision-selection evidence.	**3.1**
TBIK / Deterministic Inference [28] 2025	BF16 / FP32; matched reduction order across TP sizes	**Support; A1**: Cross-TP deterministic kernels as an alignment reference.	**3.1** (4.1; 6.3)
MiniRL [14] 2025	Hybrid-engine RL; includes FP8-rollout experiments	**Support; B1/B2**: Policy roles and mismatch-aware formulation; FP8 experiments.	**2.1** (1.2; 5.1)
Beyond Precision [18] 2026	General TIM; optimization scheduling	**Support; S**: Optimization schedule under general TIM.	**5.3** (3.3)
DPPO [29] 2026	General LLM trust-region and cross-engine mismatch evidence	**Support; B2**: Divergence-based trust regions under general TIM.	**5.2** (3.2)
VeXact / Diagnosing TIM [19] 2026	Matched-execution diagnostic reference	**Diagnostic**: Zero-mismatch diagnostic reference.	**3.4**
CIS [20] 2026	BF16 cross-engine mismatch	**Support; B1**: Confidence-dependent correction under BF16 cross-engine mismatch.	**5.1** (3.2)
Baselines and Score Centering [27] 2026	Sequence- and token-level regularized importance sampling	**Support; B1**: Granularity-dependent roles of baselines and score centering.	**5.1** (5.4)
Exploration boundary case (1)			
QeRL [2] 2025	NVFP4 with LoRA and adaptive noise	**Boundary**: NVFP4 with LoRA and adaptive noise; exploration setting.	**Box 1**

B. Source Selection and Version Handling

Literature identification. This mechanism-centered narrative review identifies contributions through targeted searches for quantized rollout, low-precision RL, and training–inference mismatch, followed by primary-text examination and backward citation tracing. Relevance follows the learner–sampler interface; numerical formats alone do not determine inclusion. The resulting collection is organized by intervention target, independently of the search terminology.

Inclusion and exclusion. Focal studies contribute a direct quantized-rollout or low-precision RL intervention, a mechanism explaining that setting, or an exploration method in which quantization changes the sampling distribution. General quantization and RL background is cited selectively. Papers devoted only to low-precision pretraining, unrelated generation modalities, or alternative rollout acceleration do not become focal entries solely because they use the same precision format.

Versions and names. The catalog separates first-publication dates, venues, and reviewed versions. QeRL, QuRL, and QaRL use formal publication records. Aliases, conference/preprint versions, and presentation changes count once. Version-unspecified arXiv links remain distinct from pinned reviewed sources because revisions can change the supported claims.

Attribution of evidence. Method claims and observations are attributed to primary sources. The likelihood, drift, and clipping identities explain mechanisms under the stated assumptions; they are distinct from experimental findings. Figures and tables are original syntheses, and constructed examples are labeled separately from published measurements. The companion evidence collection links observations to reviewed sources.

Contribution roles and empirical scope. Core, supporting, and boundary roles describe relevance to the review, independently of experimental quality. Experimental settings are recorded separately. Empirical results are taken from the cited studies and have not been independently reproduced. Training-run replication, negative results, and cross-hardware reproduction were not systematically extracted; versioned records identify the source for each reported observation.

Table 12 summarizes the reviewed settings and contribution roles.

Table 12: Coverage across independent setting axes. Joint regimes require direct evaluation.

Dimension	Represented in the focal collection	Boundary of the synthesis
Intervention	A1/A2 numerical design; B1/B2 learning rules; complementary scheduling.	Direct low-bit validation of scheduling remains limited.
Numerical interface	Weight/activation quantization, KV storage, scales, rounding, routing, and synchronization.	An interface-level result does not establish full-network agreement.
Precision	FP8, FP4/NVFP4, INT8/INT4; BF16/FP16 comparisons provide context.	Format labels omit scale granularity, precision exceptions, and native versus simulated arithmetic.
Model and adaptation	Dense and MoE models; low-rank and full-parameter updates appear in different studies.	Architecture and adaptation are separate axes; cross-regime transfer is not assumed.
Collection	Snapshot-based hybrid execution, policy lag, and quota-based partial collection.	Stateful provenance and fully overlapped execution require additional behavior and timing semantics.
Learning and generation	PPO/GRPO-style autoregressive RL; AIS also studies diffusion generation.	This survey's likelihood equations are autoregressive; denoising requires its own formulation.
Contribution role	Nineteen core studies, nine supporting studies, and one exploration boundary.	Role identifies relevance to the argument, independently of experimental strength.

C. Evidence, Assumptions, and Transfer

The behavior law depends on parameter versions, numerical execution, decoding, and retained state. Learner, proximal, and reference policies have separate roles in the loss. Execution changes can invalidate a correction's assumptions without changing the precision.

Conditional correction leaves prefix selection unchanged; truncation adds bias; alignment leaves unmatched operations; reconstructed expectations remain approximate. Transfer requires these residuals and the estimator's assumptions to be reassessed.

Conditional and complete-response correction. Equation (3) corrects actions at a fixed prefix. A response-level expectation also requires correction of how frequently that prefix is reached.

For comparison, the full-response likelihood ratio is

$$W_\theta(y \mid x) = \frac{P_\theta(y \mid x)}{Q(y \mid x)} = \prod_{t=1}^{T} \frac{p_\theta(y_t \mid h_t)}{q_{k(t)}(y_t \mid h_t)}. \tag{13}$$

Here P_θ and Q are response laws under a common stopping convention, including end-of-sequence decisions when applicable. Denominator factors must be actual conditional probabilities. Unobserved execution randomness must be marginalized or included through an extended likelihood over the corresponding execution events.

For a fixed behavior law and a parameter-independent reward R, the usual likelihood-ratio identity gives

$$\nabla_\theta \mathbb{E}_{P_\theta}[R] = \mathbb{E}_Q\left[W_\theta R \sum_{t=1}^{T} s_\theta(y_t, h_t)\right] \tag{14}$$

under support, integrability, and differentiation assumptions. The product illustrates why exact response-level correction can have high variance; multiplying only the current token ratio into an arbitrary sampled advantage is a different estimator. Nor does Equation (14) directly justify reusing group-normalized advantages computed under Q. Score Centering explicitly distinguishes sequence-level importance sampling from practical token-level corrections [10].

Figure 13 illustrates the effect of unequal prefix masses.

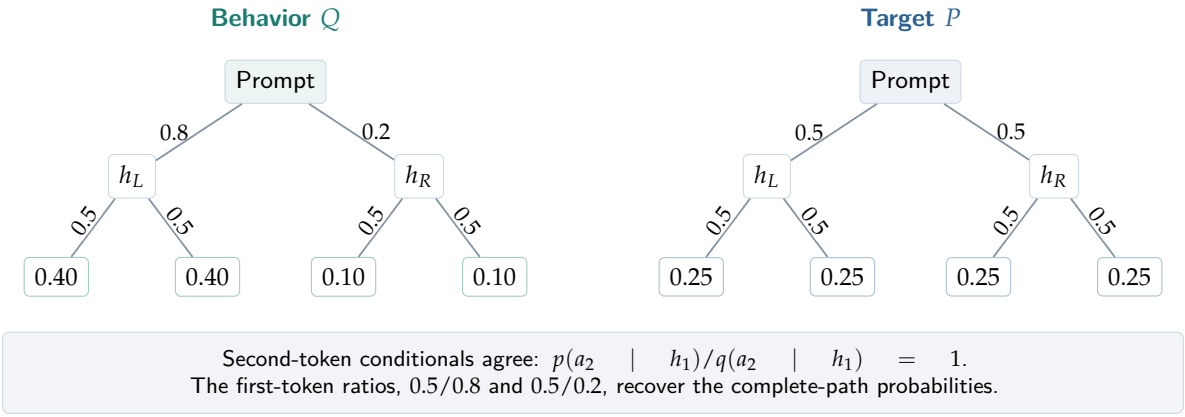


Figure 13: Conditional and response-level correction in a two-step probability tree. Leaf values are complete-response probabilities. Equal second-token conditionals coexist with different prefix masses; the product ratio restores the four target probabilities of 0.25. Constructed two-step example.

D. Terminology

Table 13: Terminology and component abbreviations used in the text and figures.

Term	Meaning
TIM	Training–inference mismatch; the survey emphasizes the learner–rollout interface.
W / A / KV	Weights, activations, and key–value cache; distinct numerical targets.
MoE / LoRA	Mixture-of-experts architecture / low-rank adaptation; architecture and adaptation are separate dimensions.
FP8 / FP4 / NVFP4	Floating-point format families with eight or four bits; NVFP4 is a particular four-bit representation. Scaling, accumulation, and exceptions also determine execution.
INT8 / INT4	Integer quantization at eight or four bits, with scales and execution semantics specified separately.
QAT	Quantization-aware training: the learner forward path models a quantized representation; this need not reproduce native kernel arithmetic.
TIS / MIS	Truncated / masked importance sampling; bounded or discarded corrections alter the estimator.
AIS / CIS	Adaptive / calibrated importance sampling: estimator-mixture adaptation and confidence-dependent truncation, respectively.
ACR / UAQ	QuRL's Adaptive Clipping Range / Update-Aware Quantization; B2 update control and A2 update visibility.
TBPO / RAC	QaRL's Trust-Band Policy Optimization / QUADS's Residual Activation Compensation; distinct objective and numerical components.
PPO / GRPO	Proximal Policy Optimization / Group Relative Policy Optimization. Their loss roles require behavior and proximal policies to be distinguished.
ESS / P3O	Effective sample size / Policy-on Policy-off Policy Optimization; batch-ratio reliability controls update influence and regularization.
DPPO / DRPO	Divergence Proximal / Divergence Regularized Policy Optimization; divergence-based masking versus smooth corrective regularization.
MIPI / MIPU	Monotonic Inference Policy Improvement / Update; inference-policy objective and practical candidate construction/acceptance.
Rollout-ResQ	Sparse residual correction of rollout activation quantization; the reviewed HiFloat4 experiments simulate FP4.
TBIK	Tree-Based Invariant Kernels; matched reduction order across tensor-parallel sizes.
Partial rollout	Collection in which a response can span collection boundaries; it does not by itself imply concurrent generation and optimization.
Cache provenance	Parameter versions and numerical execution paths that produced retained cache entries.
Behavior / proximal / reference	Actual sampling law / update-ratio anchor / KL-regularization target; see Table 1.

E. Reported Mismatch Measurements

Table 14 compares policy distances, control variables, local numerical probes, and learning outcomes across all nineteen core studies. Definitions follow each reviewed source; unspecified conventions are identified, and the companion repository links the complete records.

Table 14: Selected mismatch measurements and comparison conditions across the nineteen core studies. Source locations refer to the reviewed versions.

Study	Reported observable and location	Comparison boundary
FlashRL / TIS [1]	Cross-engine KL (Fig. 3); TIS uses sampled-action ratios.	The plotted KL direction and reduction are not specified in the caption.
Jet-RL [3]	Length- and task-conditioned learning curves (§ 3.2).	Outcome evidence; these curves do not isolate a numerical policy distance.
FP8-RL [4]	Rollout-to-training mismatch KL (§ 2.2.1; Fig. 12).	Direction is explicit; estimator and reduction require further implementation detail.
QuRL [5]	Behavior-to-proximal KL; maximum proximal/behavior ratio (Fig. 3).	Average discrepancy and rare large weights are distinct; neither is the reference KL penalty.
QaRL [6]	Proximal-ratio and mismatch-weight distributions (Fig. 4); response trust bands.	Keep the two ratios separate; geometric-mean response ratios differ from likelihood products.
P3O / ESS [21]	Valid-token normalized ESS, using current/behavior ratios (Eq. 11).	Measures concentration; includes policy change and does not identify numerical error alone.
AIS [7]	Mean absolute log gap; capped-weight ESS; advantage-weight variance amplification (§ 3.1).	These govern distinct gates; capped and raw ESS are different observables.
DRPO [22]	Sampled-action Binary-TV geometry; learning and entropy trajectories (§ 3; App. D.4).	The constraint is not full-vocabulary TV or, by itself, an isolated TIM measurement.
MIPU [23]	Sampled-token mismatch K3; inference-gap proxy and rollback rate (§ 5).	KL estimation and candidate acceptance assess different stages of an update.
QUADS [8]	Maximum log gap; layer-wise MoE L2 discrepancy (Figs. 6–7).	Local tensor error and policy-level extremes have different scales and comparison populations.
ACRL [9]	Absolute probability gap, response/group averaged relative to baseline X (Eqs. 8–12).	Distinct from log gaps; the baseline-relative controller does not define a universal tolerance.
HiFloat4 / Rollout-ResQ [24]	Residual sparsification MSE/RMSE and precision ablations (Fig. 4; § 5).	Tensor reconstruction error should not be relabeled a learner–sampler policy distance.
Miles v0.1 [25]	Same-token log-probability checks; reference-run KL; exact-mode zero difference (§ 3.1; § 5.3; Fig. 5).	Different modes use different comparisons; exact sampled-token scores do not establish full-distribution equality.
Score Centering [10]	Conditional score-mean analysis; controlled learning curves under noise, quantization, and lag (§§ 3–5).	Distinguish the analytical drift term from logged measurements and deliberately amplified stress.

Continued on next page

Table 14 continued

Study	Reported observable and location	Comparison boundary
Full Pipeline FP8 RL [11]	Probability/ratio NMAE against BF16 shadows; sign-separated over-clipping (§ 3).	Counterfactual branch errors differ from ordinary clipping rates and fixed-snapshot TIM.
QATFactory / QARL [26]	QARL learning curves and exported-policy quality (§ 4.6).	Held-out teacher KL elsewhere in the paper belongs to QAD, not an online RL TIM measurement.
TRIAGE [12]	Mean absolute gap, signed tails, directional masses, and localization (§ 3; App. B).	30B characterization is routing-conditioned; its raw magnitude is not directly comparable with ordinary 4B gaps.
TRACE [13]	Log-gap maxima/minima; relative excess train–rollout activation MSE (Figs. 4, 11).	Extremes differ from means; excess MSE uses BF16 cross-path MSE as its denominator.
KV4-RL [17]	Shared-prefix full-distribution BF16-to-quantized KL (Eq. 5.2; App. B.5).	Same checkpoint and pre-decoding probabilities isolate conditional KV distortion; trajectory shift remains outside the metric.

Measurement definitions and reproducibility. The studies differ in execution controls, reduction rules, and uncertainty units. Table 15 organizes these choices into the context needed to interpret and reproduce a reported measurement.

Table 15: Execution controls, statistical reductions, and uncertainty units used in mismatch diagnostics.

Measurement choice	Source practice or example	Required interpretation context
Execution-only comparison	KV4-RL fixes checkpoint and BF16 prefixes for full-distribution KL [17].	Prefix source; raw versus decoded probabilities; routes and cache state; KL direction.
Update-time comparison	TRIAGE replays an identical batch before/after an update with generation log-probabilities held fixed [12].	Learner versions; pre/post stage; fixed advantages and masks; route replay.
Collection-time record	CIS logs generation scores, start-of-update learner scores and rollout versions [20].	Decode versus prefill rescoring; version lag; stored numeric precision and sample selection.
Batch reliability	P3O aggregates raw-ratio moments; AIS uses truncated weights for its ESS gate [7, 21].	Weight definition, cap, valid-token count, and cross-rank reduction.
Decision disagreement	Full-pipeline FP8 compares clipping decisions with BF16 shadow ratios [11].	Reference path; advantage sign; boundary; ordinary versus counterfactual clipping.
Uncertainty and time axis	KV4-RL uses response-cluster bootstrap intervals for its paired fidelity probe [17].	Independent unit: token, response, evaluation seed, or training run; updates, sampled tokens, or elapsed time.

Policy pairs and paired execution. Learner–sampler, current–proximal, and learner–reference KL concern behavior disagreement, policy movement, and regularization. An *actor KL* label needs

its policy pair, direction, and estimator. QATFactory's teacher–student KL evaluates QAD fidelity, separately from online QARL mismatch [26].

To isolate an update's motion, replay the same prefixes before and after the update with the generation scores held fixed, as in TRIAGE's directional audit [12]. To isolate numerical execution, also match the sampling-snapshot parameters and specify kernels, routes, and cache state. A current-learner/old-behavior comparison otherwise combines execution differences and lag.

Measurement population and aggregation. Token averaging weights responses by length; response averaging gives them equal weight. For lengths 2 and 8 with constant absolute gaps 1 and 0.1, the means are 0.28 and 0.55. Global sums and counts define the token mean; P3O likewise reduces ESS moments across workers [21].

Selection can lower a diagnostic without improving agreement on the original batch. Interpretation therefore requires both a fixed valid-token population and the intervention-eligible subset, together with its occupancy and coefficient- or advantage-weighted mass. TRIAGE's separate marginal statistics, gradient-bearing occupancy, and directional masses make this distinction explicit [12]. Raw and capped ESS similarly need the weight definition and cap.

Numerical evaluation. Stored log-probability precision can impose a measurement floor. Saturation and clamping affect log-odds and binary divergences near probability one, as CIS demonstrates in a masking comparison [20]. Logit comparisons additionally require a shared offset convention, since adding the same constant to every vocabulary logit leaves all token probabilities unchanged.

Linking diagnostics to learning outcomes. An informative evaluation links the central gap and its tails to the targeted estimator or update effect, then to held-out quality and total cost. The mechanism determines the additional observable: paired activation error for rounding alignment, weight moments for estimation, or sign-conditioned branch disagreement for clipping calibration.

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